report_32bit

April 24, 2025

Introduction

There are 13 lesion biopsies in total, including AS. Different biopsy samples or even different slices from the same biopsy sample may have different amount of lesion tissues. Some completely lack lesion. List of available markers per slide:

$AS1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$AS2_Les [1] CD11c CD123 CD279 CD3 CD4 CD4_raw CD8 Foxp3 CD4T CD8T Treg

$AS3_Les [1] CD279 CD3 CD4 CD56 CD8 Foxp3 CD4T CD8T Treg

$CF_8wph [1]

$CF1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD8 Foxp3 Ki67 CD4T CD8T Treg

$CF2_Les [1] CD11c CD123 CD279 CD3 CD4 CD8 Foxp3 CD4T CD8T Treg

$CF5_Les [1] CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3 Ki67 CD4T CD8T

$CJ1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$CJ2_Les [1] CD11c CD123 CD15 CD3 CD4 CD45 CD68 CD8 CD4T CD8T

$DD_8wph [1]

$DD1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3_2 Foxp3 CD4T CD8T Treg

$DD2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$DD3_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3 CD4T CD8T Treg

$EB_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD8 Foxp3 Ki67 CD4T CD8T Treg

$JK_8wph [1]

$JK1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$JK2_Les [1] CD3 CD4 CD8 CD4T CD8T

$JK3_Les [1] CD123 CD15 CD279 CD3 CD4 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$JM1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$JM2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$JM3_Les [1] CD3 CD8 Foxp3 CD8T

$JT_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD8 Foxp3 Ki67 CD4T CD8T Treg

$LD_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$LG1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$LG2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$LG3_Les [1] CD3 CD4 CD68 CD68_raw CD8 CD4T CD8T

$LM_9wph [1]

$LM_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3 CD4T CD8T Treg

$LS_8wph [1]

$LS1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$LS2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD68_raw CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$LS3_Les [1] CD3 CD4 CD68 CD68_raw CD8 CD4T CD8T

$SK1_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD8 Foxp3 CD4T CD8T Treg

$SK2_Les [1] CD11c CD123 CD15 CD279 CD3 CD4 CD45 CD56 CD68 CD69 CD8 Foxp3 Ki67 CD4T CD8T Treg

$SK3_8wph_in_use [1]

$SK3_Les [1] CD11c CD123 CD279 CD3 CD4 CD68 CD68_raw CD8 Foxp3 CD4T CD8T Treg

The key markers are: CD3 CD11c CD123 CD15 CD56 CD68. Note that the CD68 antibody stained the epidermis of some tissues. CD68_raw.png is the raw image, and CD68.png is the processed image with reduced epidermis brightness to improve quality of segmentation. Furthermore, for CF lesion CD68 staining, CF5_Les instead of CF1_Les is used due to quality reasons.

Many lesion biopsies have multiple sections. The following choices of representative slides are made:

- AS1_Les
- CF1_Les
- CJ1_Les
- DD1_Les
- EB_Les
- JK1_Les
- JM1_Les
- JT_Les
- LD_Les
- LG2_Les
- LM_Les
- LS2_Les
- SK2_Les

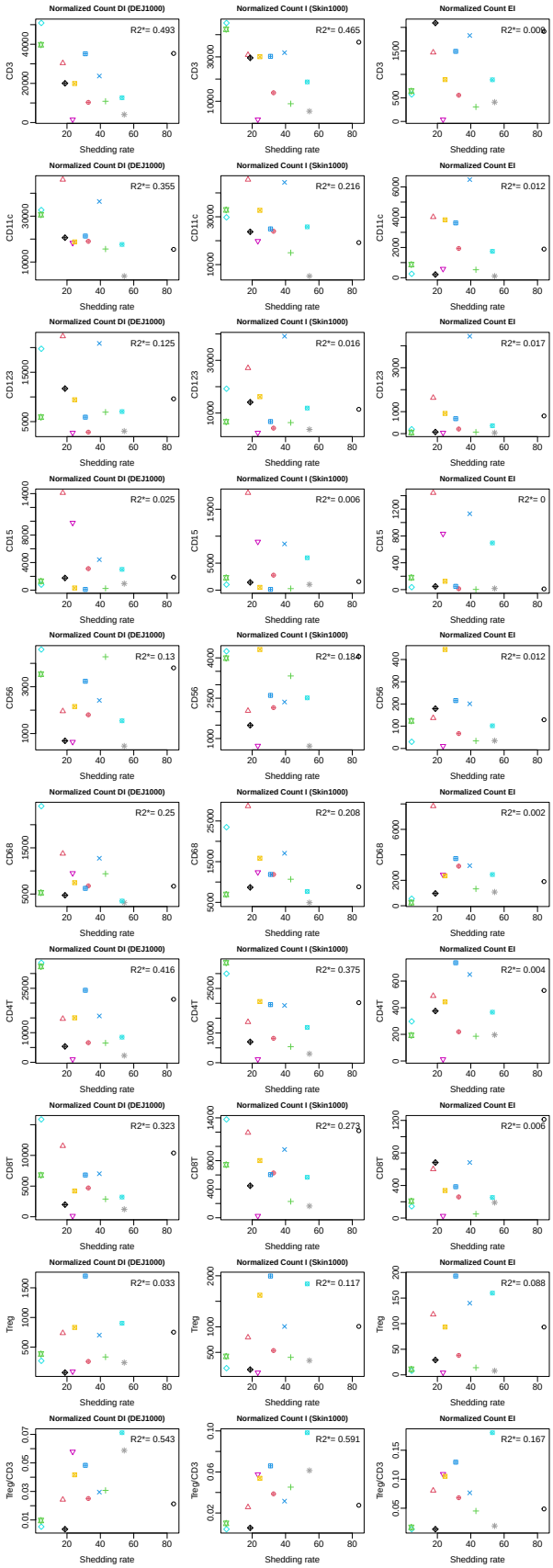
Association studies and scatterplots

Based on the previous report, we decided to

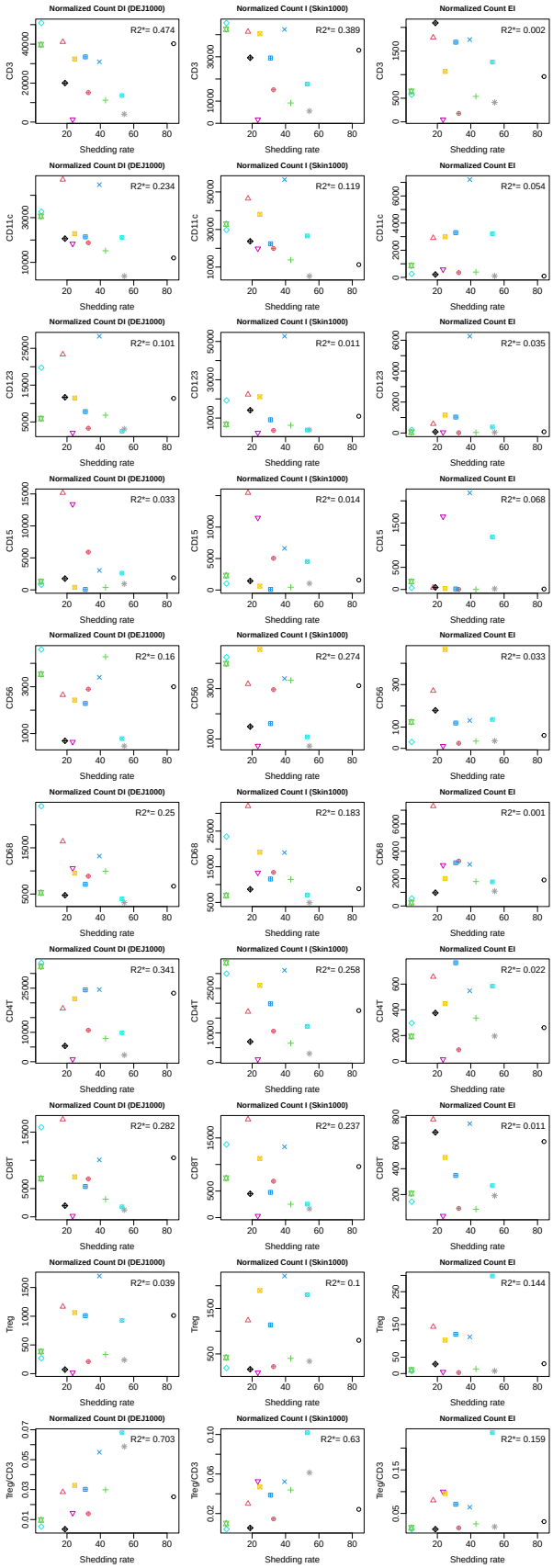
- focus on normalized densities
- focus on average across slides from the same biopsies (keep the ability for doing select slides as well)
- focus on three metrics
  - DEJ1000, DI
  - skin1000, I
  - EI
- expand to additional markers
- expand to examine Treg, which are CD4+CD8+Foxp3+

Normalized count is the number of cells in a vertical strip 100um wide.

R2* is computed without slides from AS_Les, which has the highest shedding rate and appears an outlier.



Each biopsy is represented by the average of all available slides from the biopsy.



Each biopsy is represented by one select slide.

Table 1: CD3, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	1915	35300	0	0	37215
CF_Les_avg	1467	30358	1661	26907	60393
CJ_Les_avg	310	10863	0	0	11172
DD_Les_avg	1828	23766	711	15273	41577
EB_Les_avg	575	50879	0	0	51454
JK_Les_avg	41	1555	0	0	1596
JM_Les_avg	890	19945	305	14413	35554
JT_Les_avg	410	4067	0	0	4477
LD_Les_avg	2094	20040	0	0	22134
LG_Les_avg	558	10305	529	9289	20681
LM_Les_avg	646	39738	0	0	40384
LS_Les_avg	1491	35143	0	0	36633
SK_Les_avg	886	12675	630	1876	16067

Table 2: CD3, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	36633	0	36633
CF_Les_avg	30886	26914	57800
CJ_Les_avg	8894	0	8894
DD_Les_avg	31896	17519	49415
EB_Les_avg	45277	0	45277
JK_Les_avg	1804	0	1804
JM_Les_avg	30095	14337	44432
JT_Les_avg	5519	0	5519
LD_Les_avg	29572	0	29572
LG_Les_avg	13847	10977	24824
LM_Les_avg	42423	0	42423
LS_Les_avg	30250	0	30250
SK_Les_avg	18723	4020	22743

Table 3: CD11c, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	1892	15498	0	0	17390
CF_Les_avg	4012	45940	10158	38850	98960
CJ_Les_avg	528	15666	0	0	16194
DD_Les_avg	6484	36398	30813	40444	114139
EB_Les_avg	257	32673	0	0	32930
JK_Les_avg	585	18349	0	0	18934
JM_Les_avg	3823	18764	3646	23314	49548
JT_Les_avg	107	3843	0	0	3950
LD_Les_avg	212	20605	0	0	20817
LG_Les_avg	1934	19026	3772	18664	43396
LM_Les_avg	871	30612	0	0	31483
LS_Les_avg	3622	21322	0	0	24945
SK_Les_avg	1744	17653	4535	5925	29858

Table 4: CD11c, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	19204	0	19204
CF_Les_avg	45563	40808	86371
CJ_Les_avg	14960	0	14960
DD_Les_avg	44405	61945	106350
EB_Les_avg	29777	0	29777
JK_Les_avg	19849	0	19849
JM_Les_avg	32713	33248	65962
JT_Les_avg	5181	0	5181
LD_Les_avg	23725	0	23725
LG_Les_avg	23958	26404	50362
LM_Les_avg	32881	0	32881
LS_Les_avg	24928	0	24928
SK_Les_avg	25785	13177	38962

Table 5: CD123, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	806	9609	0	0	10414
CF_Les_avg	1631	22295	1846	8533	34304
CJ_Les_avg	71	6913	0	0	6984
DD_Les_avg	4433	20840	6276	27489	59038
EB_Les_avg	199	19758	0	0	19957
JK_Les_avg	40	2742	0	0	2782
JM_Les_avg	920	9420	1783	17052	29174
JT_Les_avg	40	3075	0	0	3115
LD_Les_avg	74	11693	0	0	11767
LG_Les_avg	210	2865	2218	2858	8152
LM_Les_avg	49	5907	0	0	5956
LS_Les_avg	682	5902	0	0	6584
SK_Les_avg	360	7034	1603	2948	11945

Table 6: CD123, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	11412	0	11412
CF_Les_avg	27073	11148	38222
CJ_Les_avg	6386	0	6386
DD_Les_avg	39136	45100	84236
EB_Les_avg	19228	0	19228
JK_Les_avg	2533	0	2533
JM_Les_avg	16224	25126	41350
JT_Les_avg	3797	0	3797
LD_Les_avg	14110	0	14110
LG_Les_avg	4281	9620	13900
LM_Les_avg	6703	0	6703
LS_Les_avg	6788	0	6788
SK_Les_avg	11892	7610	19501

Table 7: CD15, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	10	1887	0	0	1897
CF_Les_avg	1442	14110	5018	1462	22032
CJ_Les_avg	4	238	0	0	242
DD_Les_avg	1131	4418	32161	5974	43683
EB_Les_avg	36	793	0	0	829
JK_Les_avg	832	9765	0	0	10598
JM_Les_avg	125	308	1156	718	2308
JT_Les_avg	17	943	0	0	960
LD_Les_avg	48	1762	0	0	1810
LG_Les_avg	13	3112	72	150	3347
LM_Les_avg	179	1292	0	0	1471
LS_Les_avg	49	104	0	0	153
SK_Les_avg	696	3009	7526	2510	13740

Table 8: CD15, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	1583	0	1583
CF_Les_avg	18028	5978	24006
CJ_Les_avg	276	0	276
DD_Les_avg	8567	38625	47192
EB_Les_avg	1026	0	1026
JK_Les_avg	8964	0	8964
JM_Les_avg	510	2294	2805
JT_Les_avg	1053	0	1053
LD_Les_avg	1437	0	1437
LG_Les_avg	2769	284	3053
LM_Les_avg	2291	0	2291
LS_Les_avg	135	0	135
SK_Les_avg	6000	8188	14187

Table 9: CD56, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	129	3806	0	0	3934
CF_Les_avg	137	1956	126	1738	3958
CJ_Les_avg	34	4280	0	0	4314
DD_Les_avg	201	2417	68	1835	4521
EB_Les_avg	30	4592	0	0	4622
JK_Les_avg	11	643	0	0	654
JM_Les_avg	445	2154	120	1989	4707
JT_Les_avg	35	467	0	0	502
LD_Les_avg	179	689	0	0	868
LG_Les_avg	67	1800	20	1011	2898
LM_Les_avg	124	3538	0	0	3662
LS_Les_avg	216	3236	0	0	3452
SK_Les_avg	102	1546	46	668	2362

Table 10: CD56, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	4050	0	4050
CF_Les_avg	2031	1936	3967
CJ_Les_avg	3327	0	3327
DD_Les_avg	2359	1404	3763
EB_Les_avg	4245	0	4245
JK_Les_avg	730	0	730
JM_Les_avg	4311	2240	6552
JT_Les_avg	722	0	722
LD_Les_avg	1492	0	1492
LG_Les_avg	2149	1098	3246
LM_Les_avg	3989	0	3989
LS_Les_avg	2605	0	2605
SK_Les_avg	2513	892	3405

Table 11: CD68, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	1900	6687	0	0	8587
CF_Les_avg	7826	13743	0	0	21569
CJ_Les_avg	1326	9383	0	0	10708
DD_Les_avg	3148	12738	4807	18167	38860
EB_Les_avg	551	24026	0	0	24577
JK_Les_avg	2432	9506	0	0	11938
JM_Les_avg	2350	7454	3658	10738	24201
JT_Les_avg	1080	3126	0	0	4206
LD_Les_avg	968	4728	0	0	5696
LG_Les_avg	3112	6754	3898	3086	16850
LM_Les_avg	243	5272	0	0	5515
LS_Les_avg	3709	6225	0	0	9934
SK_Les_avg	2452	3482	0	0	5934

Table 12: CD68, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	8813	0	8813
CF_Les_avg	28655	0	28655
CJ_Les_avg	10684	0	10684
DD_Les_avg	17060	23578	40638
EB_Les_avg	23472	0	23472
JK_Les_avg	12394	0	12394
JM_Les_avg	15849	19716	35564
JT_Les_avg	4920	0	4920
LD_Les_avg	8679	0	8679
LG_Les_avg	11854	10274	22128
LM_Les_avg	6937	0	6937
LS_Les_avg	11862	0	11862
SK_Les_avg	7665	0	7665

Table 13: CD4T, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	529	21283	0	0	21813
CF_Les_avg	488	14689	361	12359	27896
CJ_Les_avg	186	6524	0	0	6709
DD_Les_avg	649	15652	244	9954	26500
EB_Les_avg	297	33639	0	0	33936
JK_Les_avg	13	1006	0	0	1019
JM_Les_avg	444	15026	146	14012	29628
JT_Les_avg	196	2267	0	0	2463
LD_Les_avg	375	5366	0	0	5741
LG_Les_avg	219	6614	164	5461	12458
LM_Les_avg	193	32370	0	0	32563
LS_Les_avg	737	24341	0	0	25078
SK_Les_avg	366	8480	207	1214	10267

Table 14: CD4T, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	20206	0	20206
CF_Les_avg	13697	9407	23104
CJ_Les_avg	5400	0	5400
DD_Les_avg	19277	9339	28616
EB_Les_avg	29975	0	29975
JK_Les_avg	1153	0	1153
JM_Les_avg	20588	13644	34231
JT_Les_avg	3014	0	3014
LD_Les_avg	7020	0	7020
LG_Les_avg	8204	6029	14234
LM_Les_avg	33641	0	33641
LS_Les_avg	19562	0	19562
SK_Les_avg	11888	2292	14179

Table 15: CD8T, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	1212	10388	0	0	11601
CF_Les_avg	601	11540	823	16175	29140
CJ_Les_avg	50	2887	0	0	2938
DD_Les_avg	683	7013	185	4515	12396
EB_Les_avg	144	15863	0	0	16007
JK_Les_avg	26	139	0	0	165
JM_Les_avg	338	4208	76	3483	8105
JT_Les_avg	191	1225	0	0	1416
LD_Les_avg	682	1979	0	0	2661
LG_Les_avg	259	4693	218	4044	9215
LM_Les_avg	208	6790	0	0	6998
LS_Les_avg	385	6803	0	0	7188
SK_Les_avg	252	3210	207	1214	4883

Table 16: CD8T, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	12194	0	12194
CF_Les_avg	11915	15902	27818
CJ_Les_avg	2284	0	2284
DD_Les_avg	9563	4361	13924
EB_Les_avg	13781	0	13781
JK_Les_avg	259	0	259
JM_Les_avg	8015	3843	11857
JT_Les_avg	1645	0	1645
LD_Les_avg	4482	0	4482
LG_Les_avg	6276	4706	10983
LM_Les_avg	7428	0	7428
LS_Les_avg	6047	0	6047
SK_Les_avg	5673	2292	7964

Table 17: Treg, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	93	752	0	0	845
CF_Les_avg	118	737	90	1423	2368
CJ_Les_avg	14	334	0	0	348
DD_Les_avg	140	702	26	963	1831
EB_Les_avg	8	269	0	0	277
JK_Les_avg	4	90	0	0	94
JM_Les_avg	94	832	32	594	1551
JT_Les_avg	8	239	0	0	247
LD_Les_avg	29	71	0	0	100
LG_Les_avg	38	258	16	248	560
LM_Les_avg	11	384	0	0	395
LS_Les_avg	193	1698	0	0	1892
SK_Les_avg	160	903	57	59	1178

Table 18: Treg, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	1008	0	1008
CF_Les_avg	792	1238	2030
CJ_Les_avg	402	0	402
DD_Les_avg	1007	809	1816
EB_Les_avg	190	0	190
JK_Les_avg	104	0	104
JM_Les_avg	1620	818	2437
JT_Les_avg	339	0	339
LD_Les_avg	162	0	162
LG_Les_avg	535	339	874
LM_Les_avg	421	0	421
LS_Les_avg	1995	0	1995
SK_Les_avg	1841	349	2189

Table 19: Treg/CD3, DEJ1000, slides average. Normalized counts in different regions: dermis intact (DI); dermis ulcerative (DU); epidermis intact (EI); epidermis ulcerative (EU). Total = row sum.

	EI	DI	EU	DU	total
AS_Les_avg	0	0	NaN	NaN	NaN
CF_Les_avg	0	0	0	0	0
CJ_Les_avg	0	0	NaN	NaN	NaN
DD_Les_avg	0	0	0	0	0
EB_Les_avg	0	0	NaN	NaN	NaN
JK_Les_avg	0	0	NaN	NaN	NaN
JM_Les_avg	0	0	0	0	0
JT_Les_avg	0	0	NaN	NaN	NaN
LD_Les_avg	0	0	NaN	NaN	NaN
LG_Les_avg	0	0	0	0	0
LM_Les_avg	0	0	NaN	NaN	NaN
LS_Les_avg	0	0	NaN	NaN	NaN
SK_Les_avg	0	0	0	0	0

Table 20: Treg/CD3, Skin1000, slides average. Normalized counts in different regions: intact (I); ulcerative (U). Total = row sum.

	I	U	total
AS_Les_avg	0	NaN	NaN
CF_Les_avg	0	0	0
CJ_Les_avg	0	NaN	NaN
DD_Les_avg	0	0	0
EB_Les_avg	0	NaN	NaN
JK_Les_avg	0	NaN	NaN
JM_Les_avg	0	0	0
JT_Les_avg	0	NaN	NaN
LD_Les_avg	0	NaN	NaN
LG_Les_avg	0	0	0
LM_Les_avg	0	NaN	NaN
LS_Les_avg	0	NaN	NaN
SK_Les_avg	0	0	0

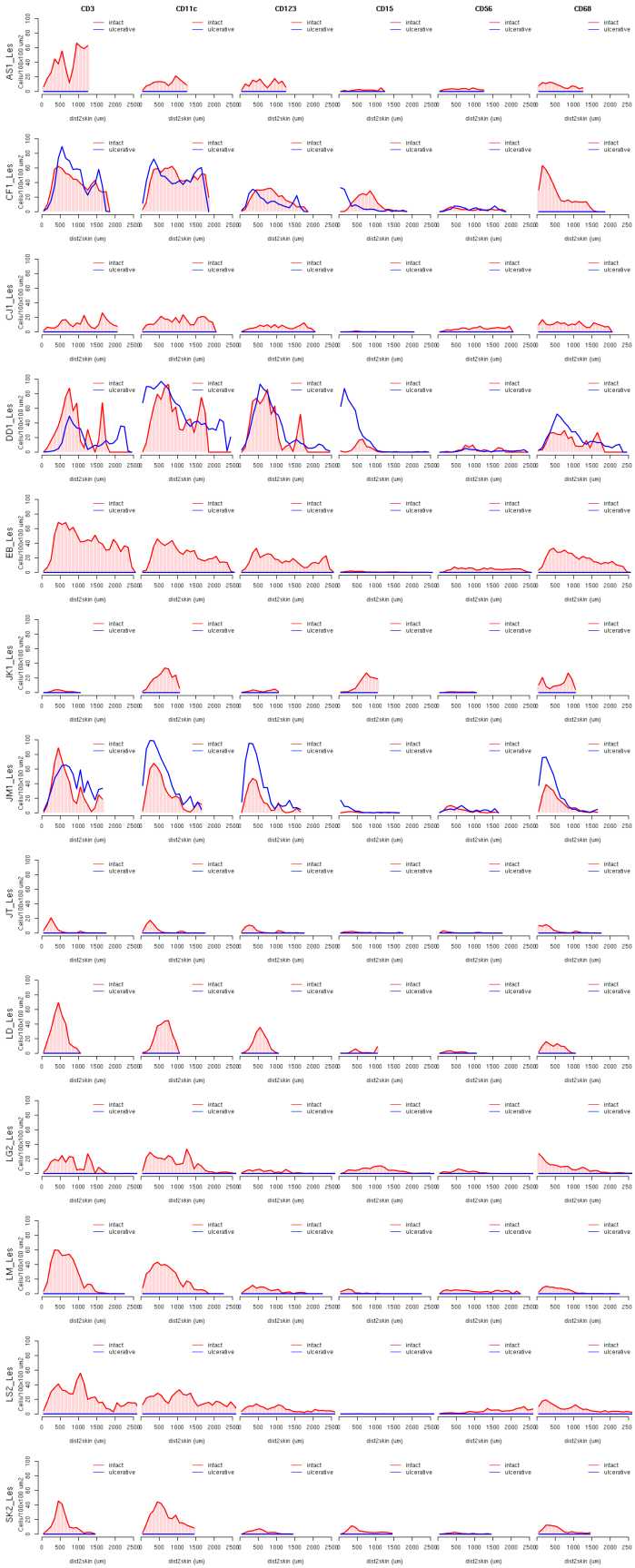
Density curves

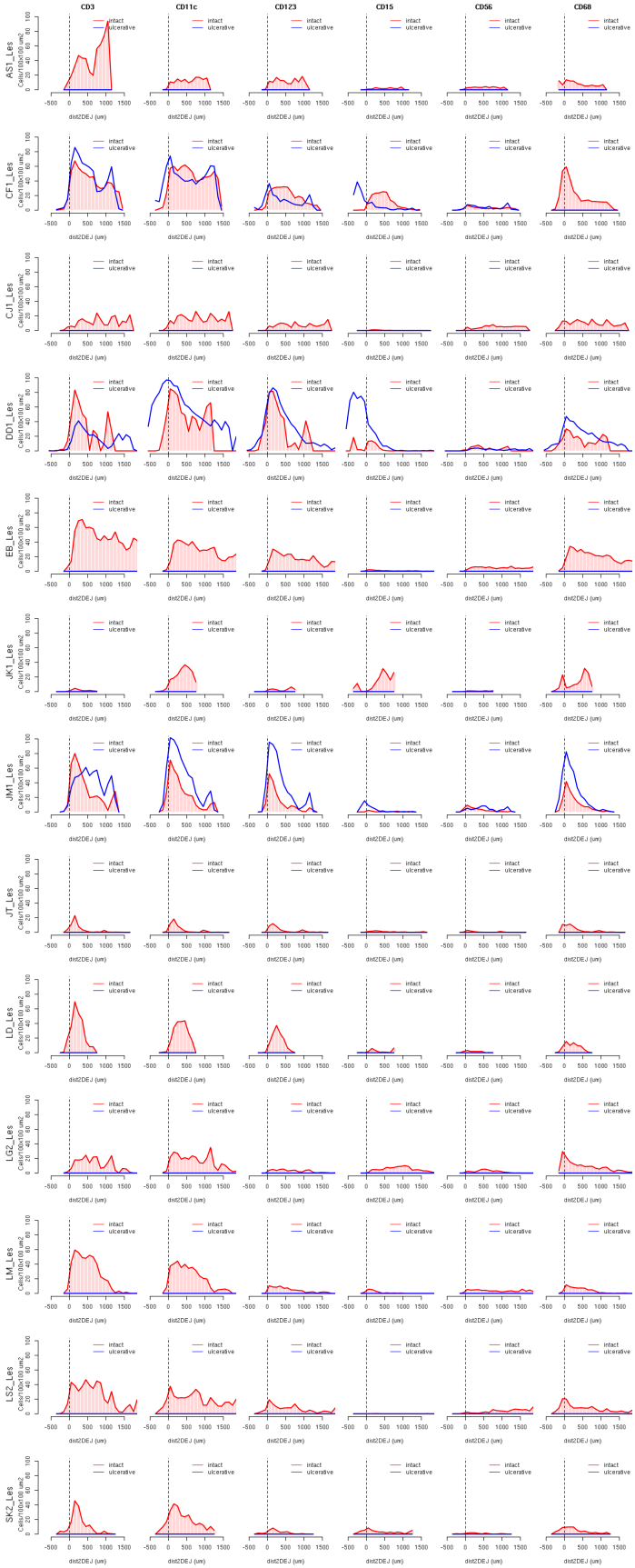
In this section we plot the number of cells per 100x100 um2 tissue as a function of distance to skin or DEJ. Each bin in the histogram represents the count in a 100x100 um2 area. Normalized density used in the correlates study is computed as the area under the curve with cutoff at a given depth.

Intact vs. ulcerative - representative slides

In this subsection, there are two figures, one for dist2skin and one for dist2DEJ. Each figure shows 13 representative slides. Some observations:

- Comparing the CD3 cell distribution between CF and DD, we see that the ratio of the area under the curve across both epi and dermis between red and blue(ulcerative and intact) is about 1 for CD3/CF and 1/3 for CD3/DD. This suggests that CD3 is depleted in lesion in DD, but not in CF.
- Comparing the CD15 cell distribution in CF between intact and ulcerative, we see that the lesion peak is shifted to the left compared to the normal peak. This pattern is more clear in CF than in DD. We don't see this pattern for CD3.

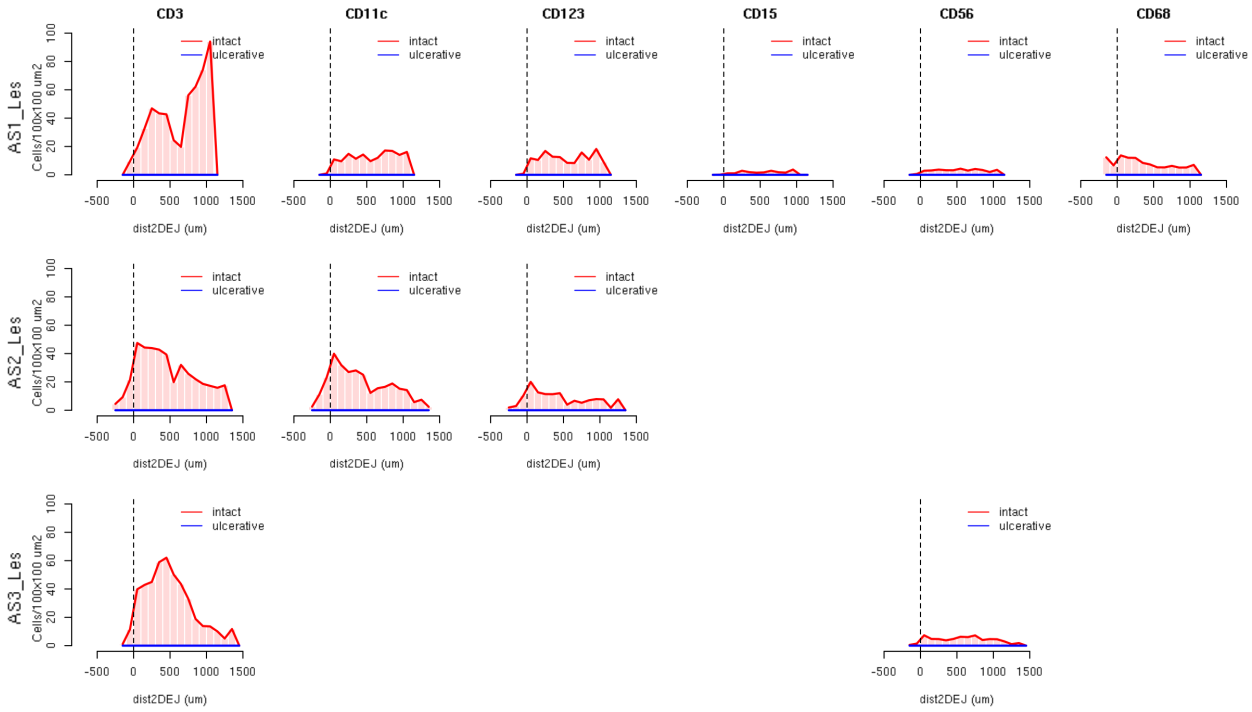

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100x100 um2
dist2skin (um)

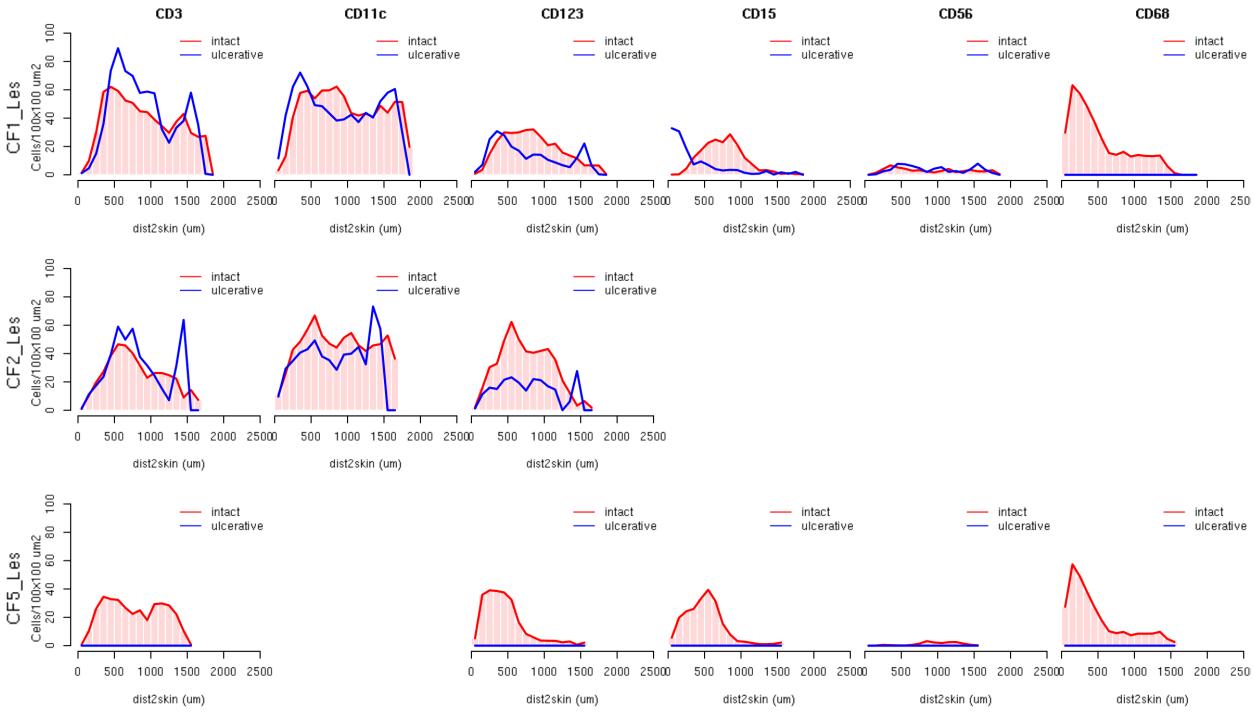

CD3
CD11c
CD123
CD15
CD56
CD68
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100x100 um2
dist2DEJ (um)
intact
ulcerative

Intact vs. ulcerrative - all slides

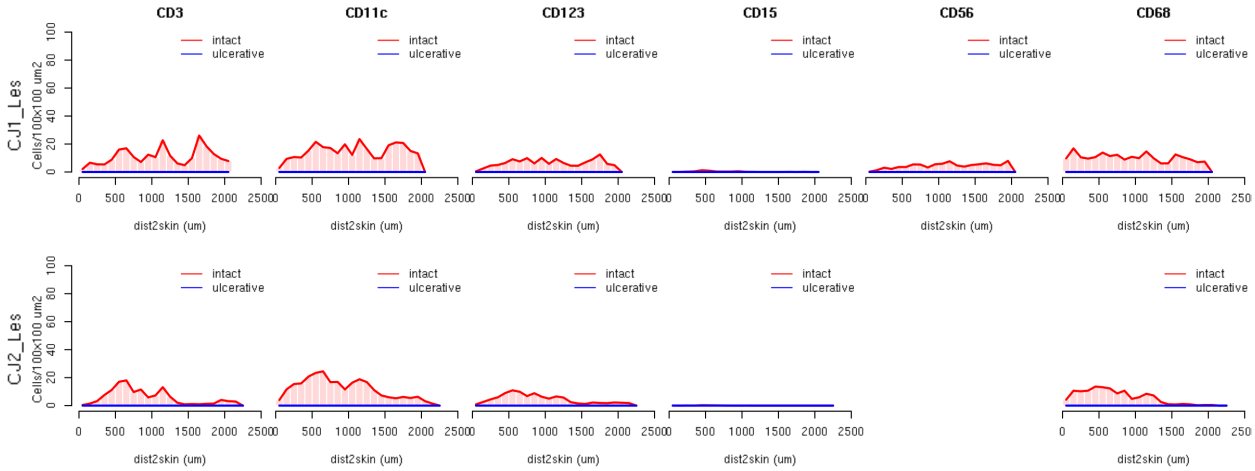
In this subsection, each figure shows all slides from a single biopsy.

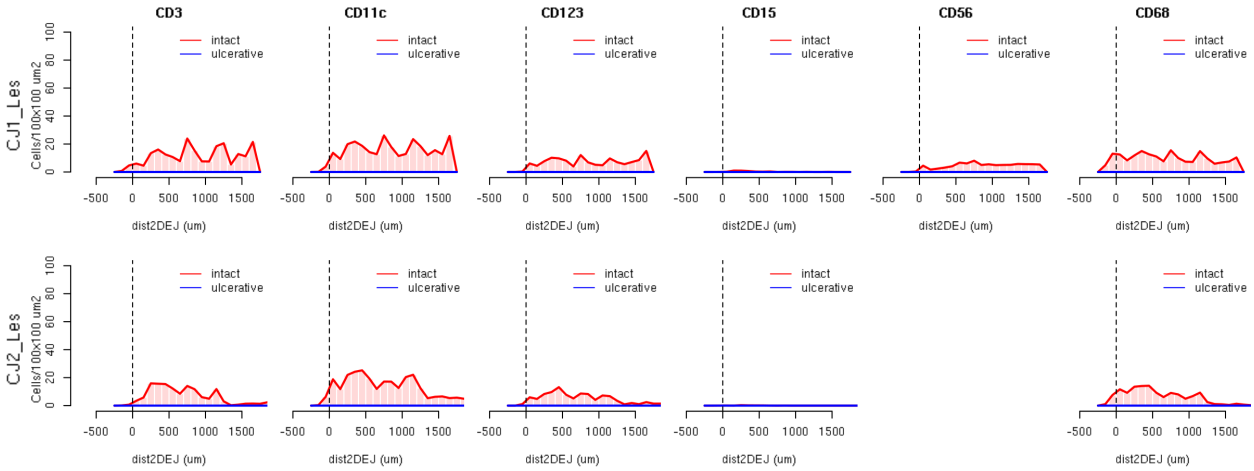

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
dist2skin (um)

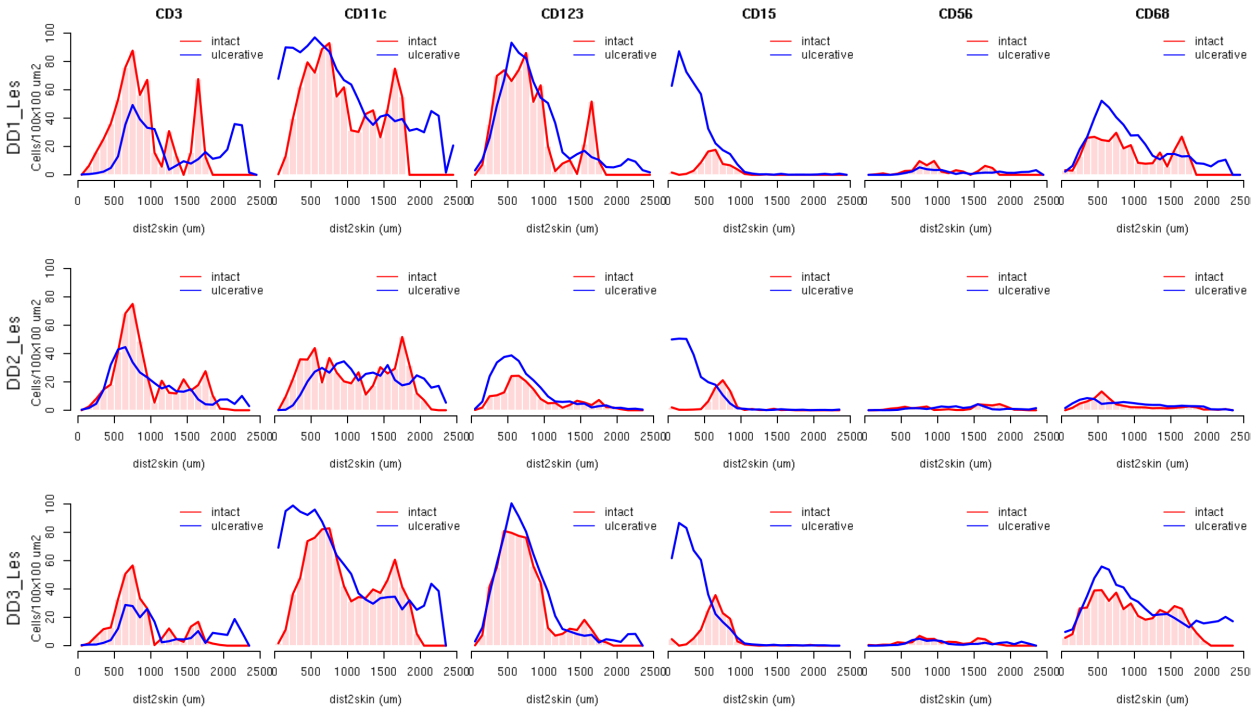

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

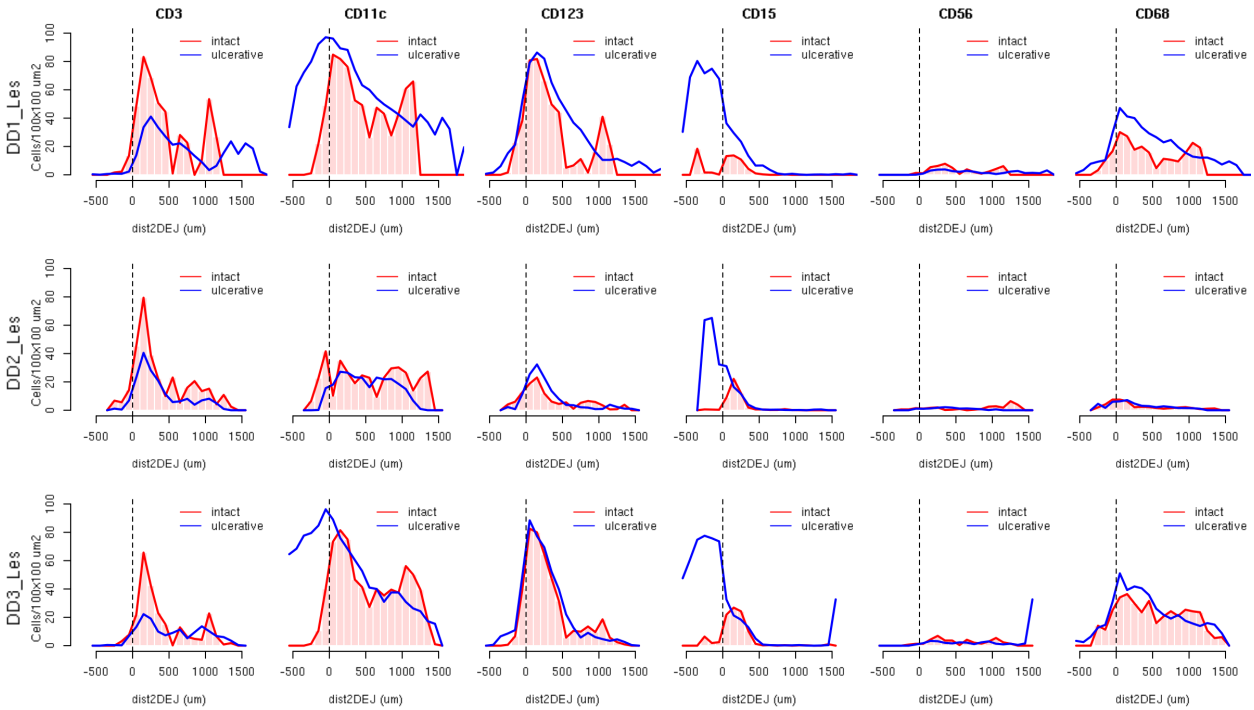

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
dist2skin (um)

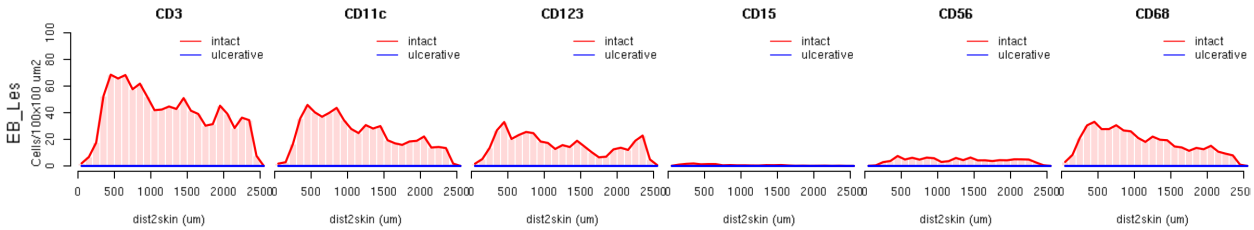

CD3
CD11c
CD123
CD15
CD56
CD68
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
intact
ulcerative
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100

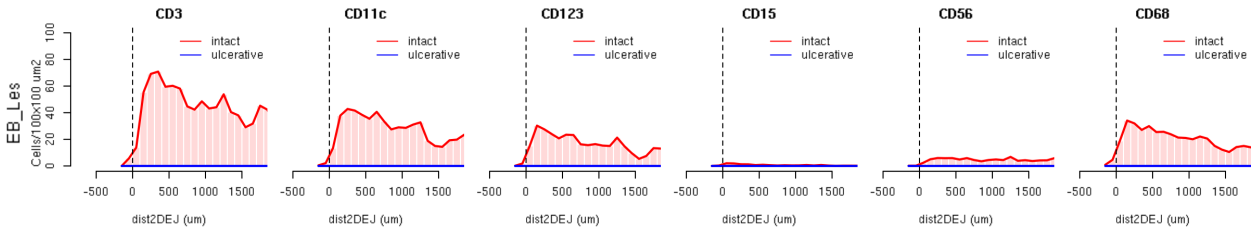

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
dist2skin (um)

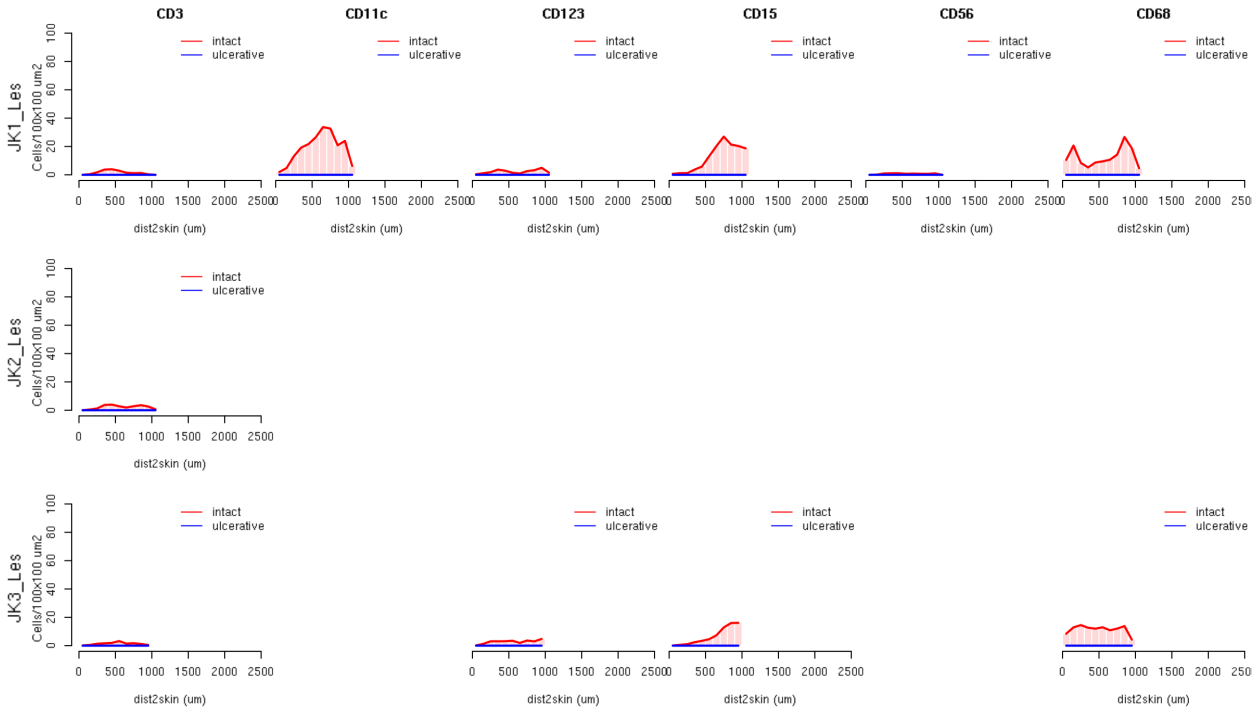

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

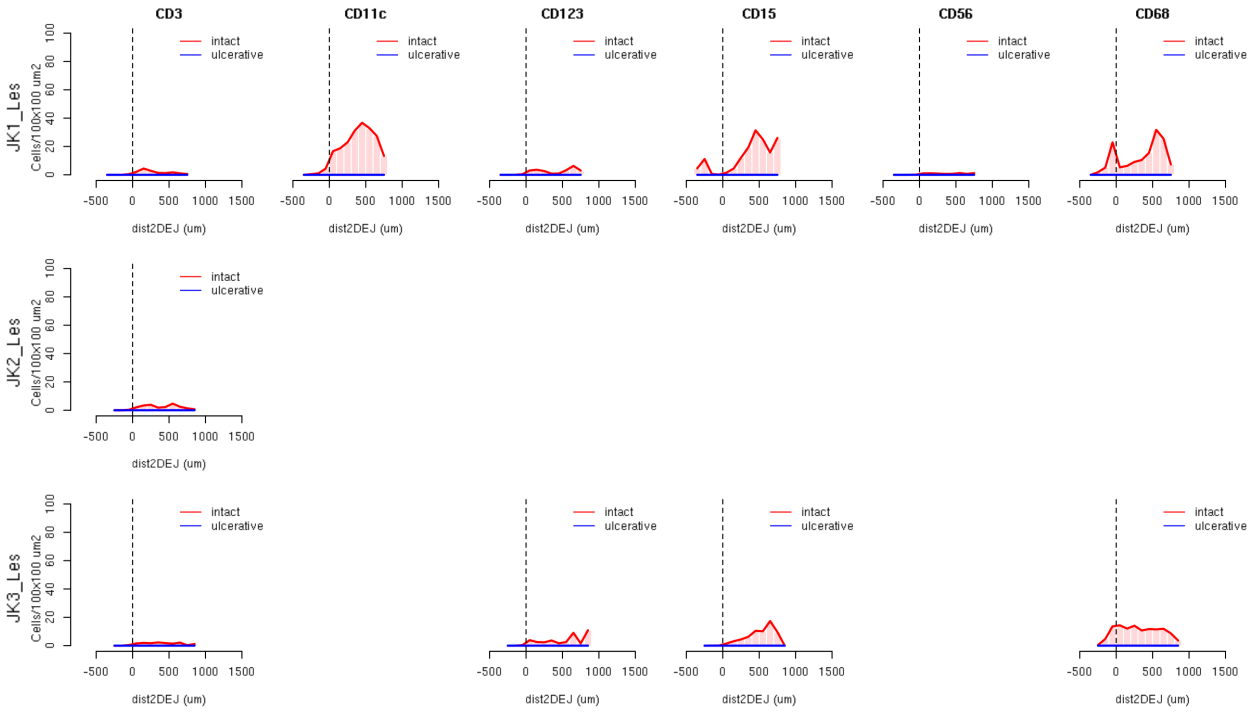

CD3
CD11c
CD123
CD15
CD56
CD68
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
intact
ulcerative
dist2skin (um)

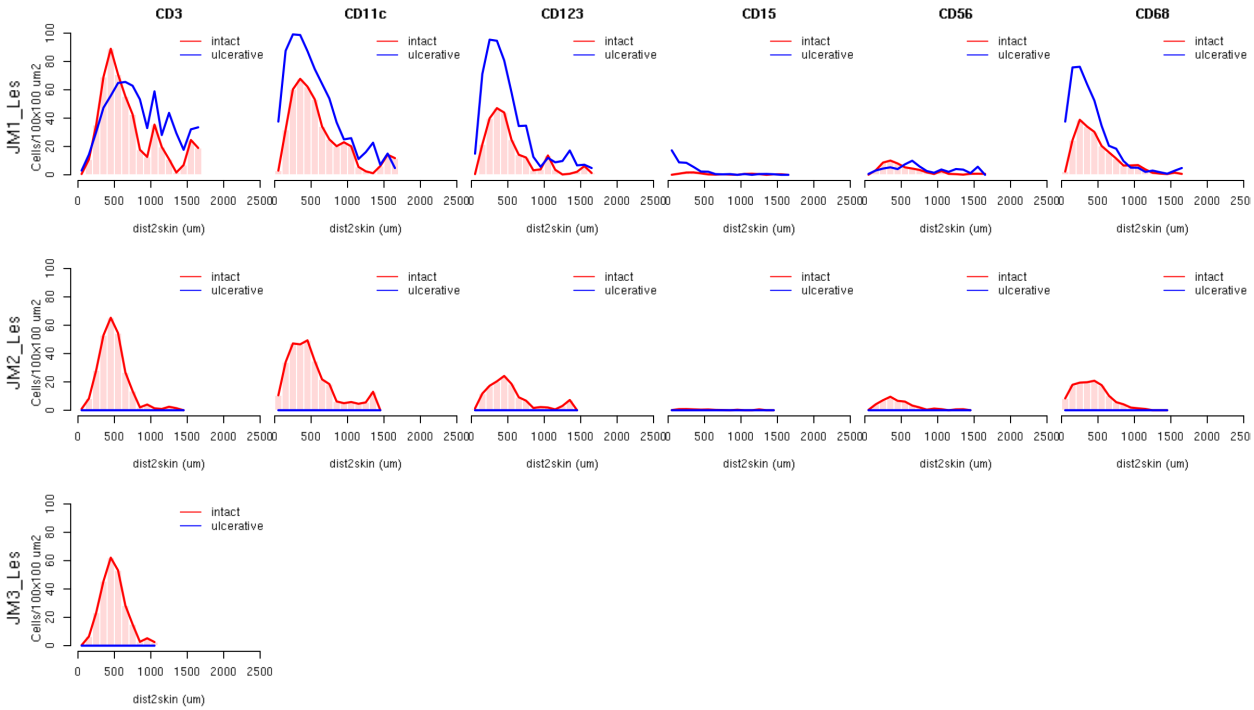

CD3
CD11c
CD123
CD15
CD56
CD68
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
intact
ulcerative
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100

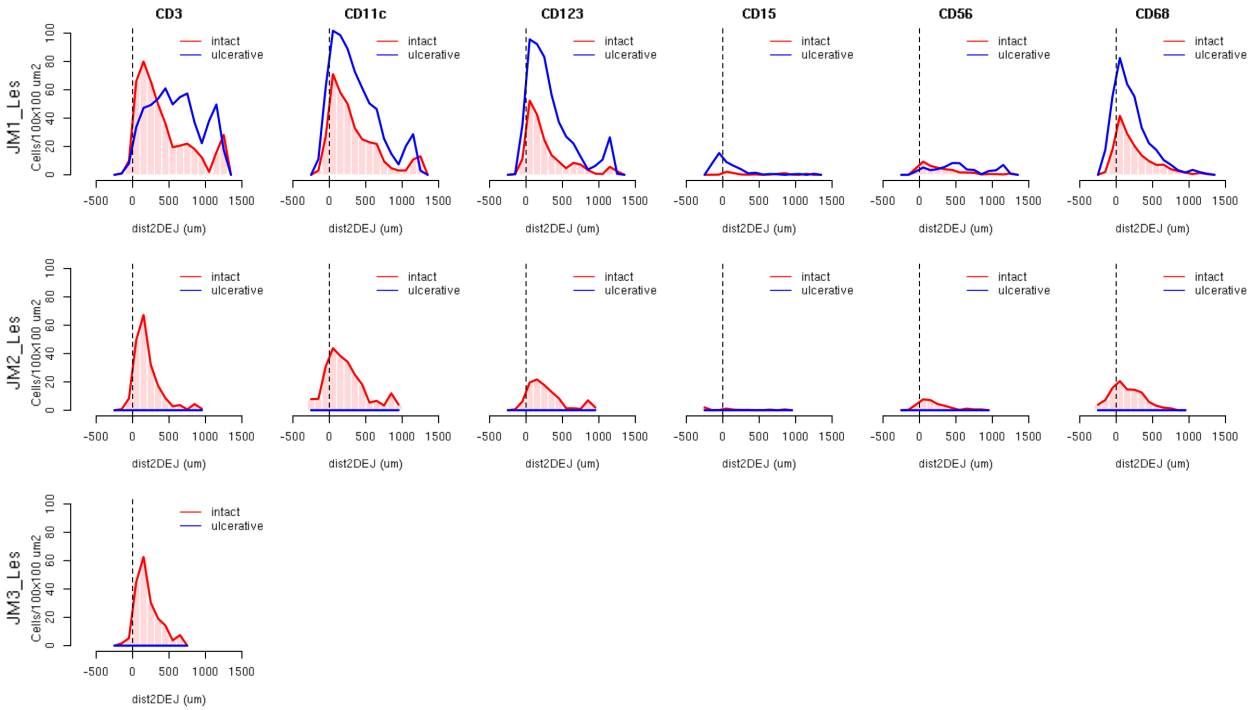

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
EB_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)

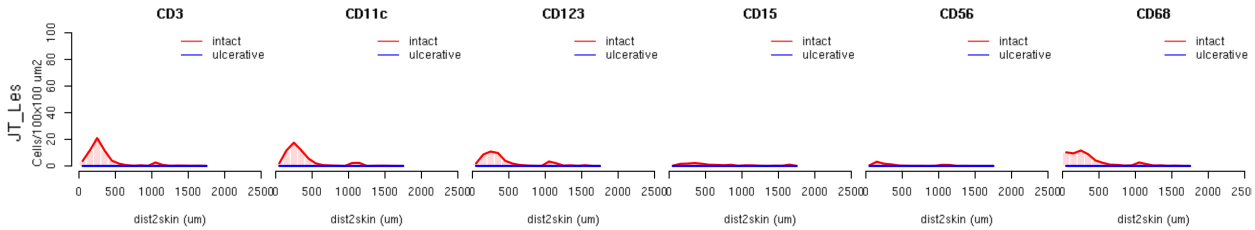

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
EB_Les
Cells/100x100 um2
0 20 40 60 80 100
-500 0 500 1000 1500
dist2DEJ (um)

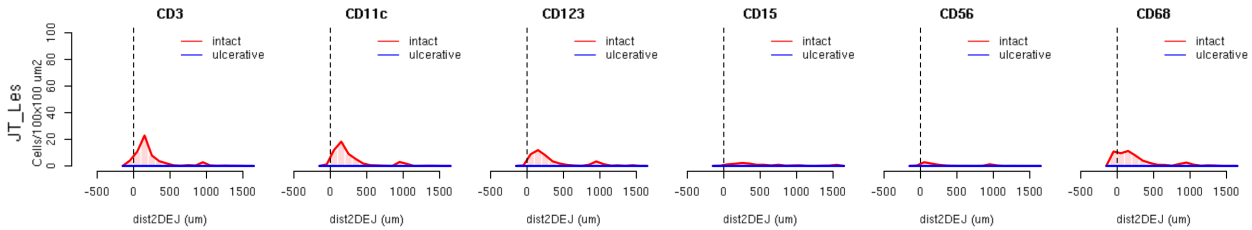

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
dist2skin (um)

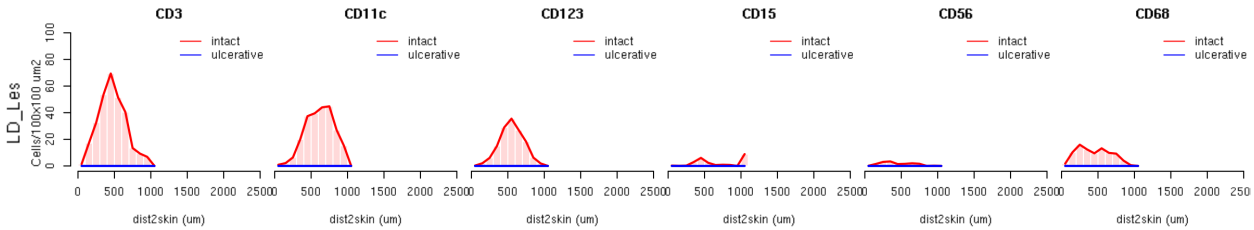

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100

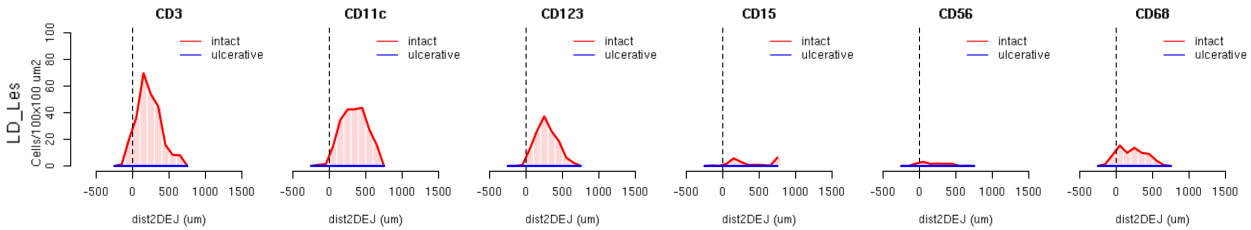

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
dist2skin (um)

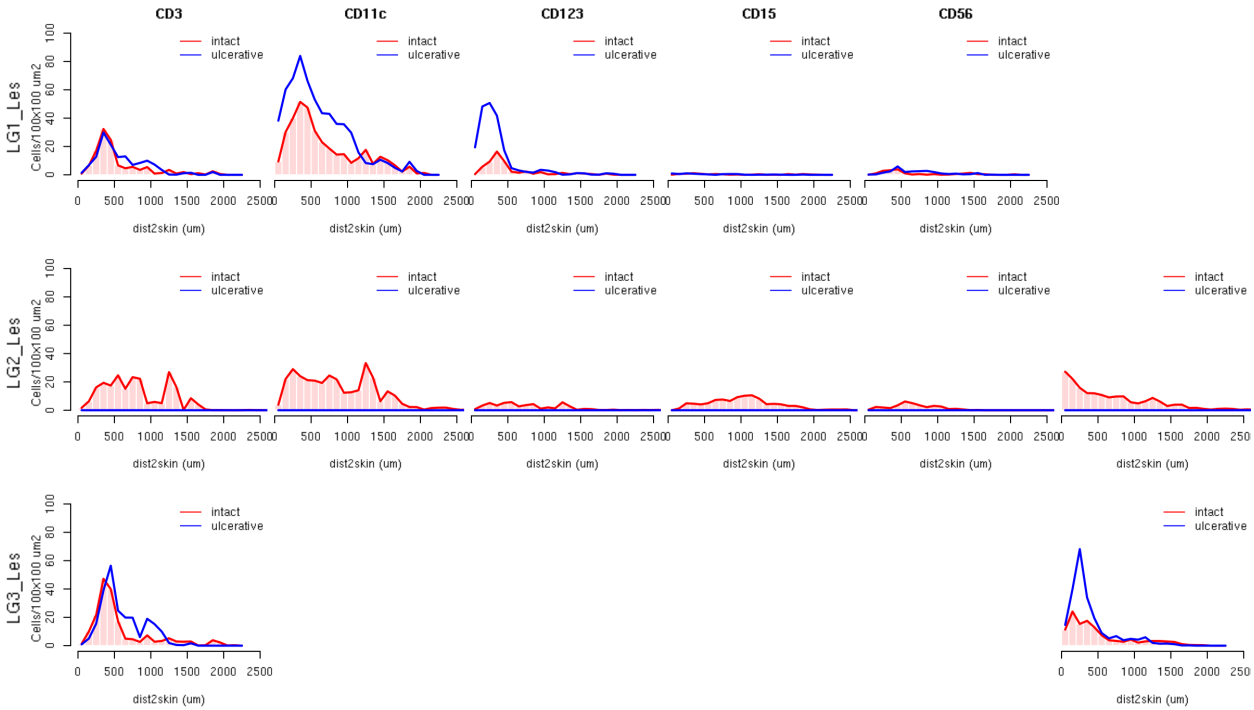

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

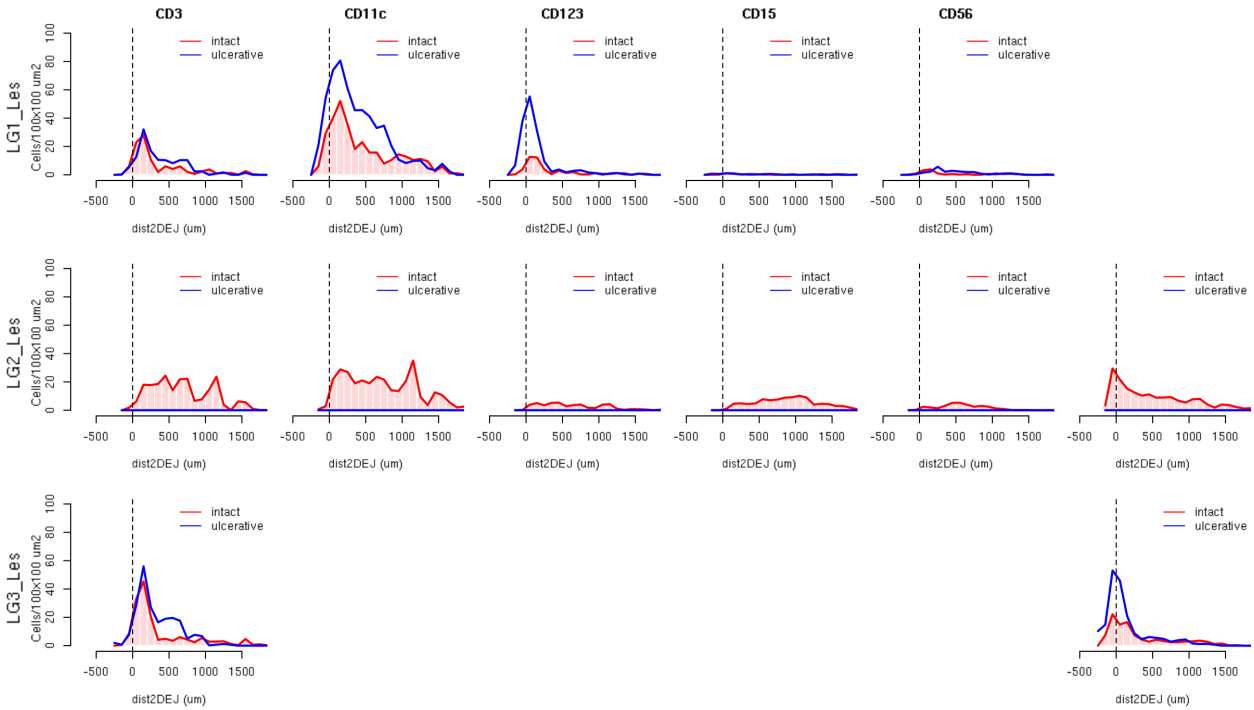

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JT_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)

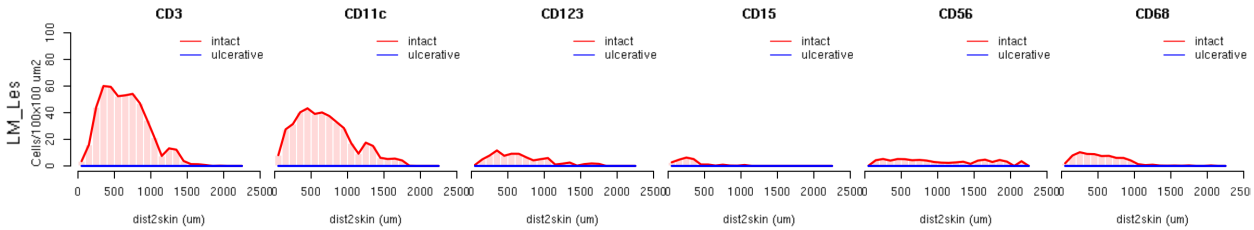

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
JT_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

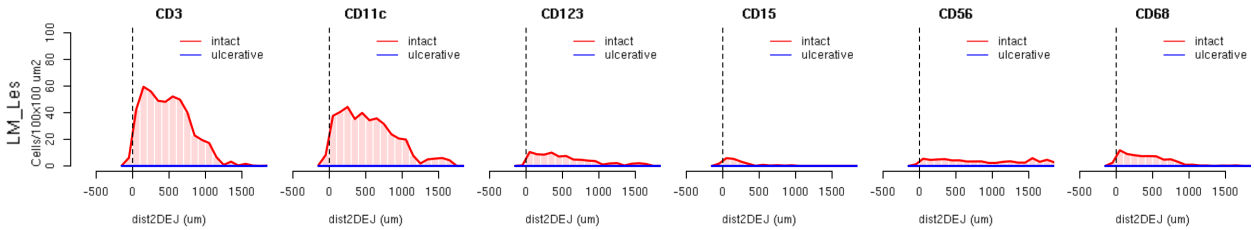

CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LD_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LD_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

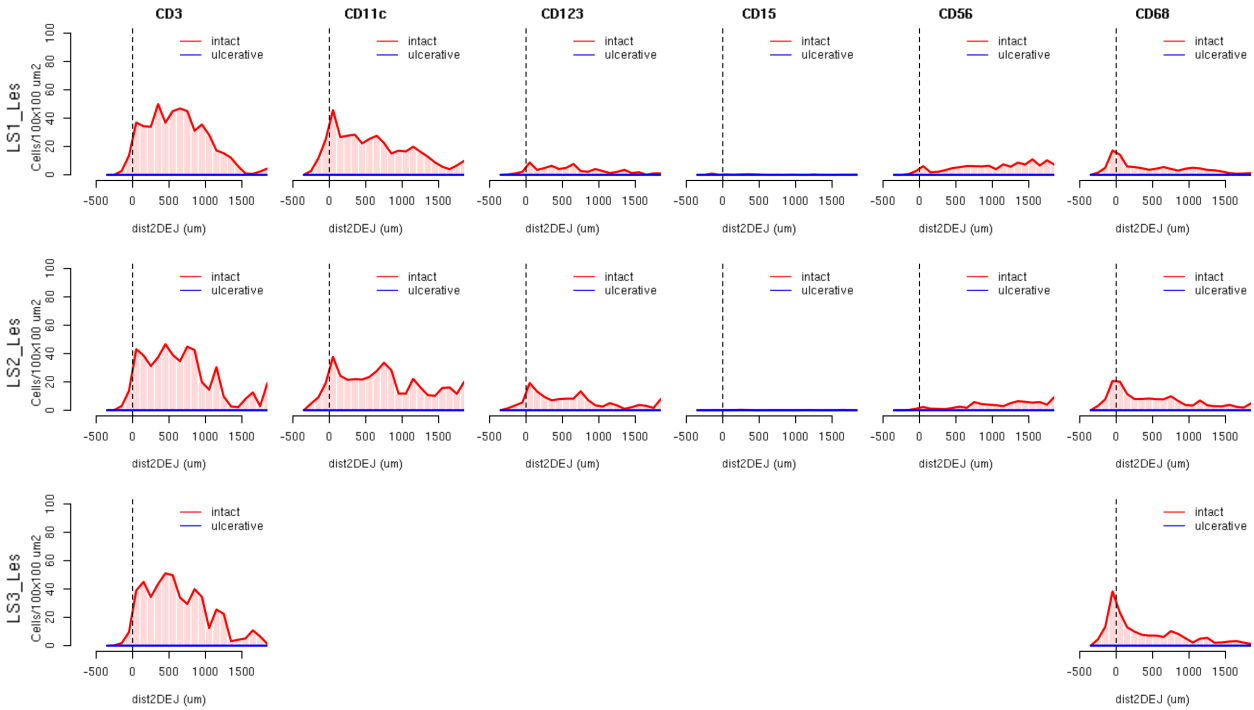

CD3
CD11c
CD123
CD15
CD56
intact
ulcerative
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
dist2skin (um)

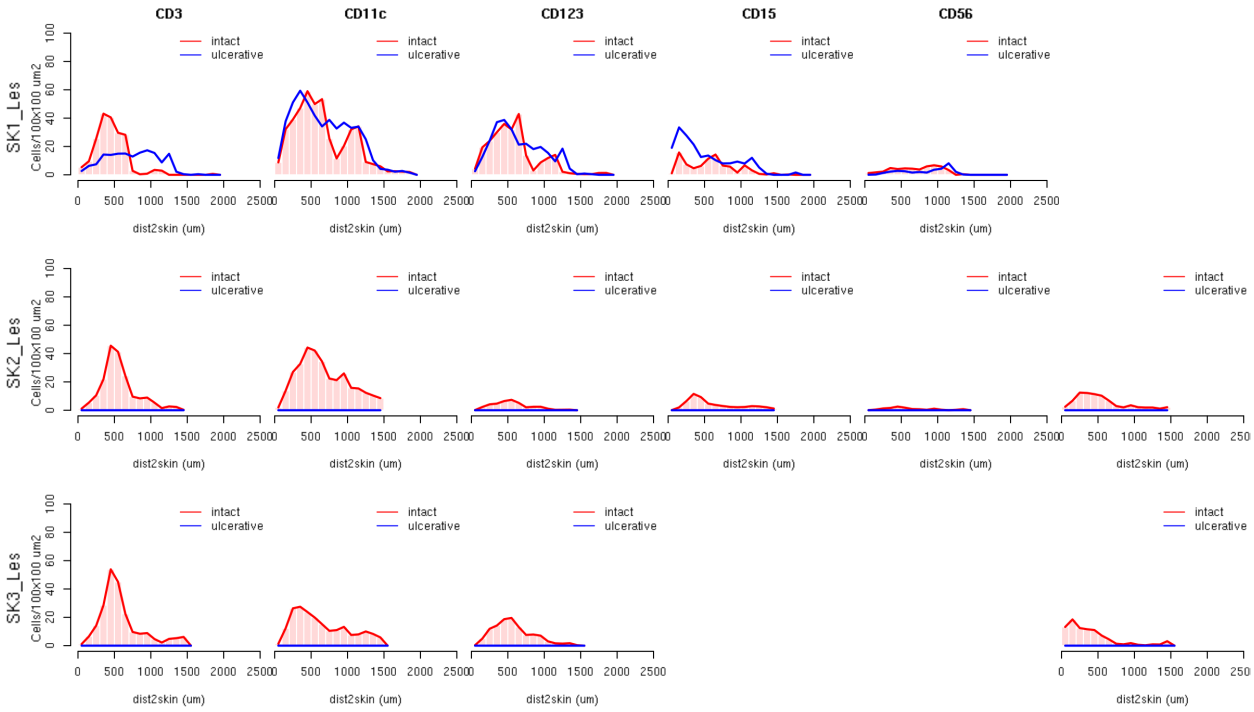

CD3
CD11c
CD123
CD15
CD56
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
intact
ulcerative
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LM_Les
Cells/100x100 um2
0
20
40
60
80
100
0
500
1000
1500
2000
2500
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LM_Les
Cells/100x100 um2
0 20 40 60 80 100
-500 0 500 1000 1500
dist2DEJ (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
CD68
intact
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
0
20
40
60
80
100
-500
0
500
1000
1500
dist2DEJ (um)

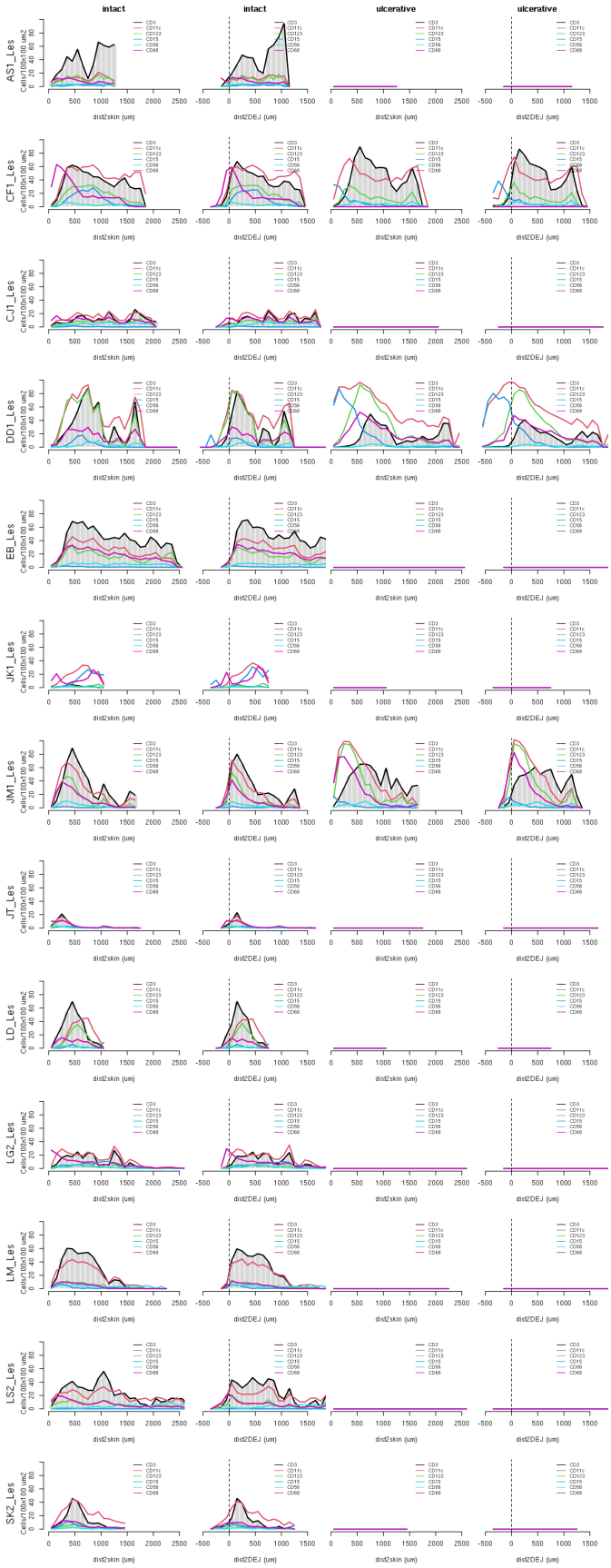

CD3
CD11c
CD123
CD15
CD56
intact
ulcerative
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
dist2skin (um)


CD3
CD11c
CD123
CD15
CD56
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
intact
ulcerative
dist2DEJ (um)
-500
0
500
1000
1500
0
20
40
60
80
100

Markers comparison - select slides

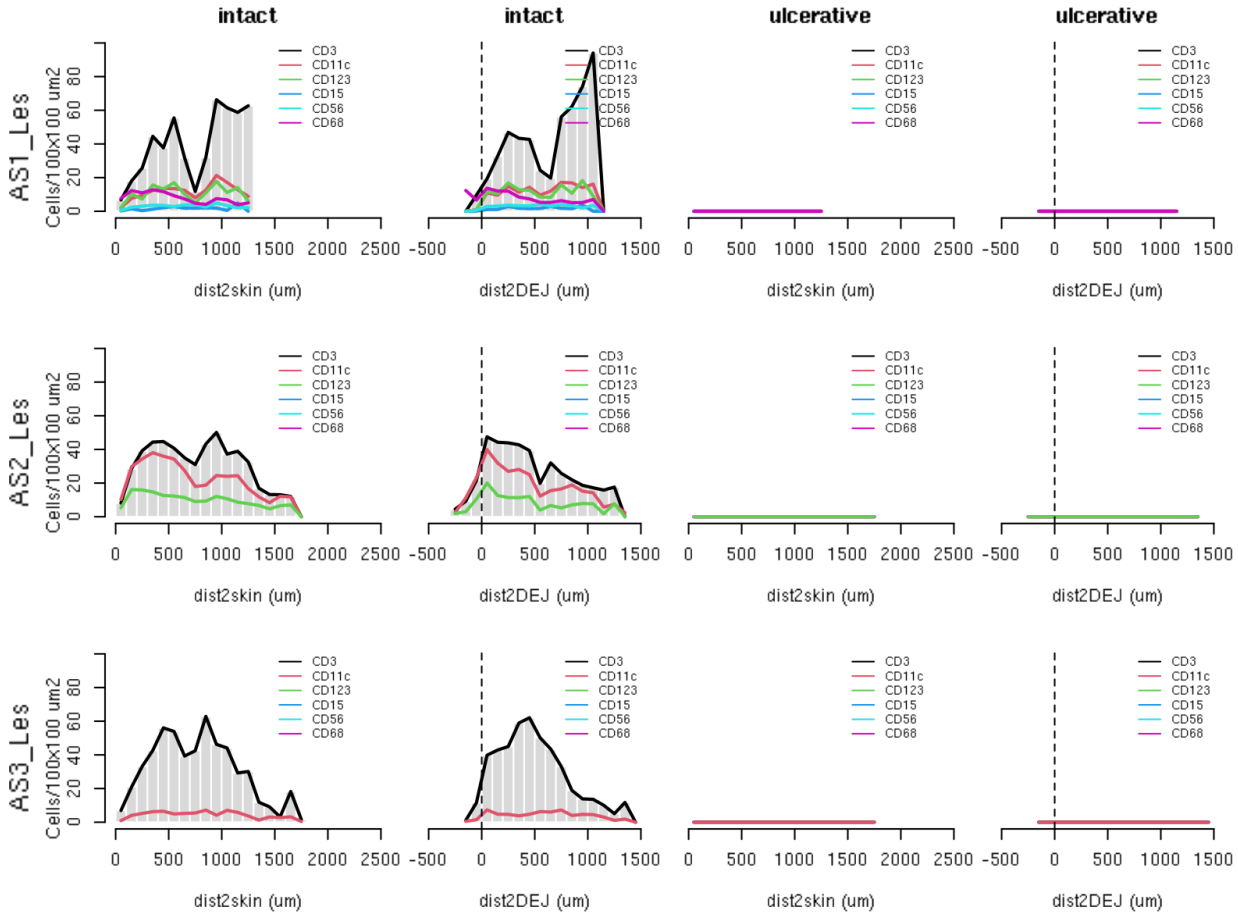
In this subsection, there is one figure showing 13 representative slides. Some observations:

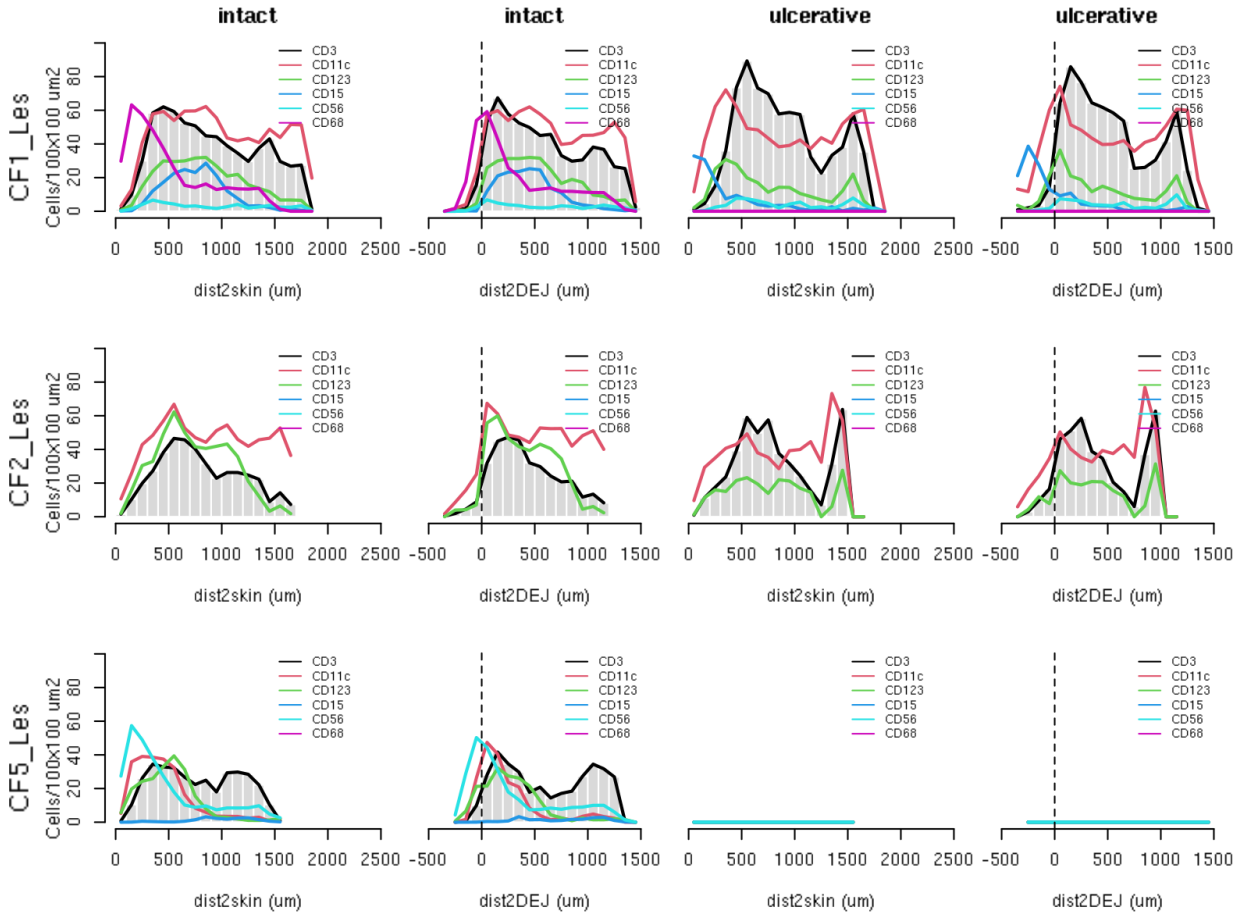
- The CD15 dist2skin peak in DD/lesion is near skin while the corresponding dist2DEJ peak is near DEJ is due to the fact that the DEJ line is very curvy.

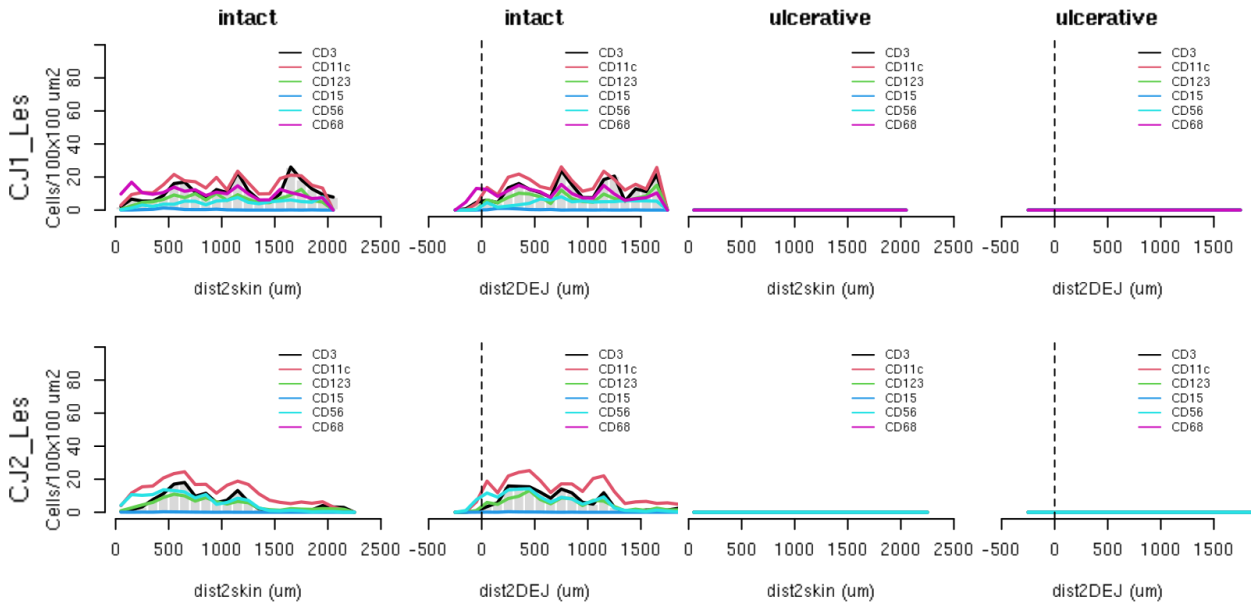

intact
intact
ulcerative
ulcerative
AS1_Les
CF1_Les
CJ1_Les
DD1_Les
EB_Les
JK1_Les
JM1_Les
JT_Les
LD_Les
LG2_Les
LM_Les
LS2_Les
SK2_Les
Cells/100um^2
dist2skin (um)
dist2DEJ (um)
CD3
CD11c
CD123
CD15
CD56
CD68

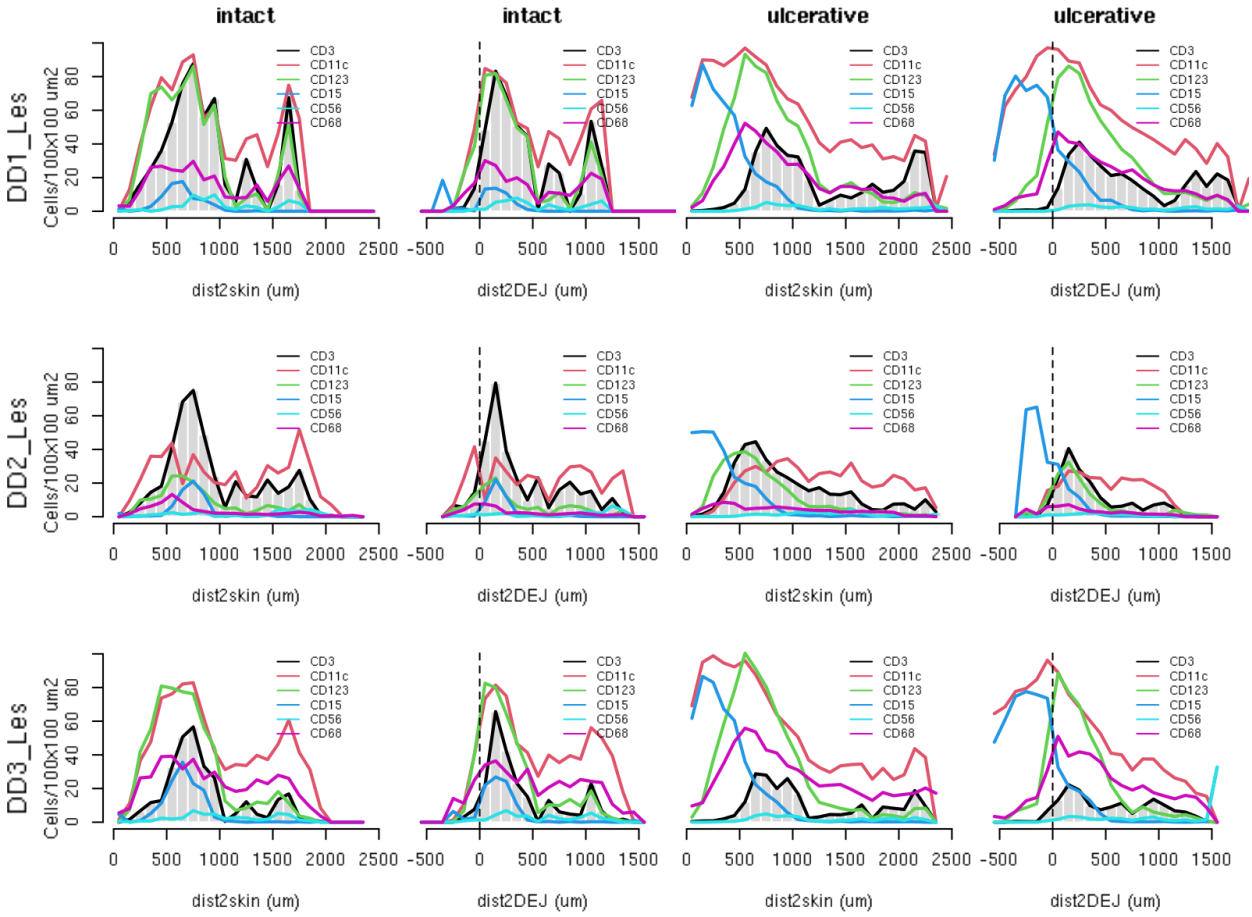
Markers comparison - all slides

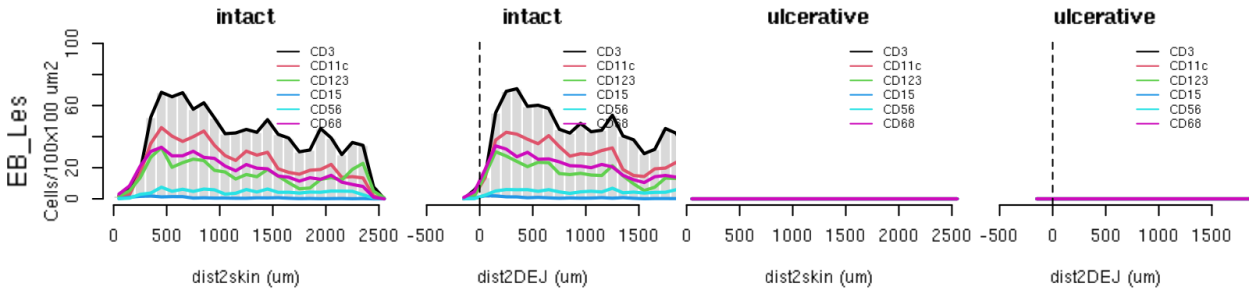
In this subsection, each figure shows all slides from a single biopsy.

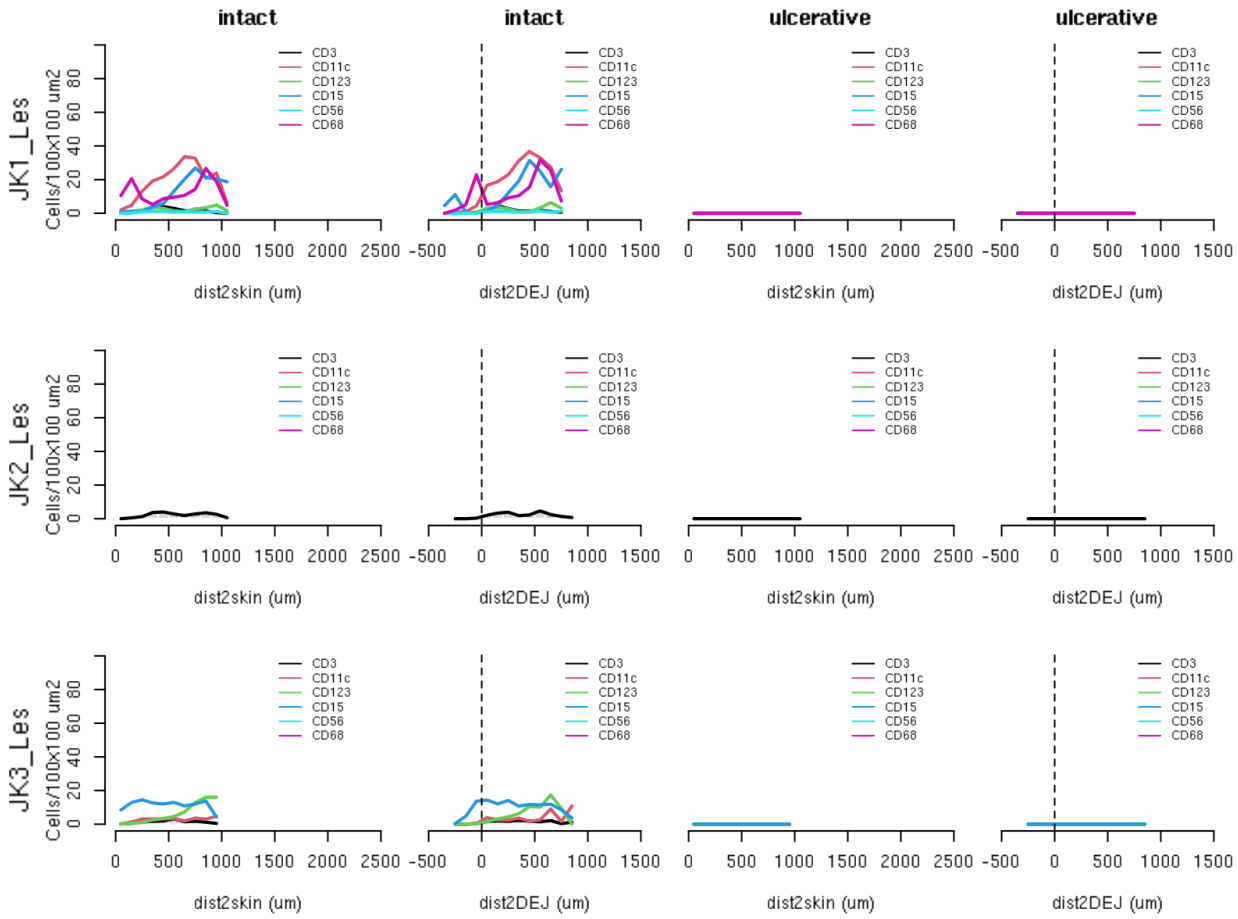

intact
intact
ulcerative
ulcerative
AS1_Les
AS2_Les
AS3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

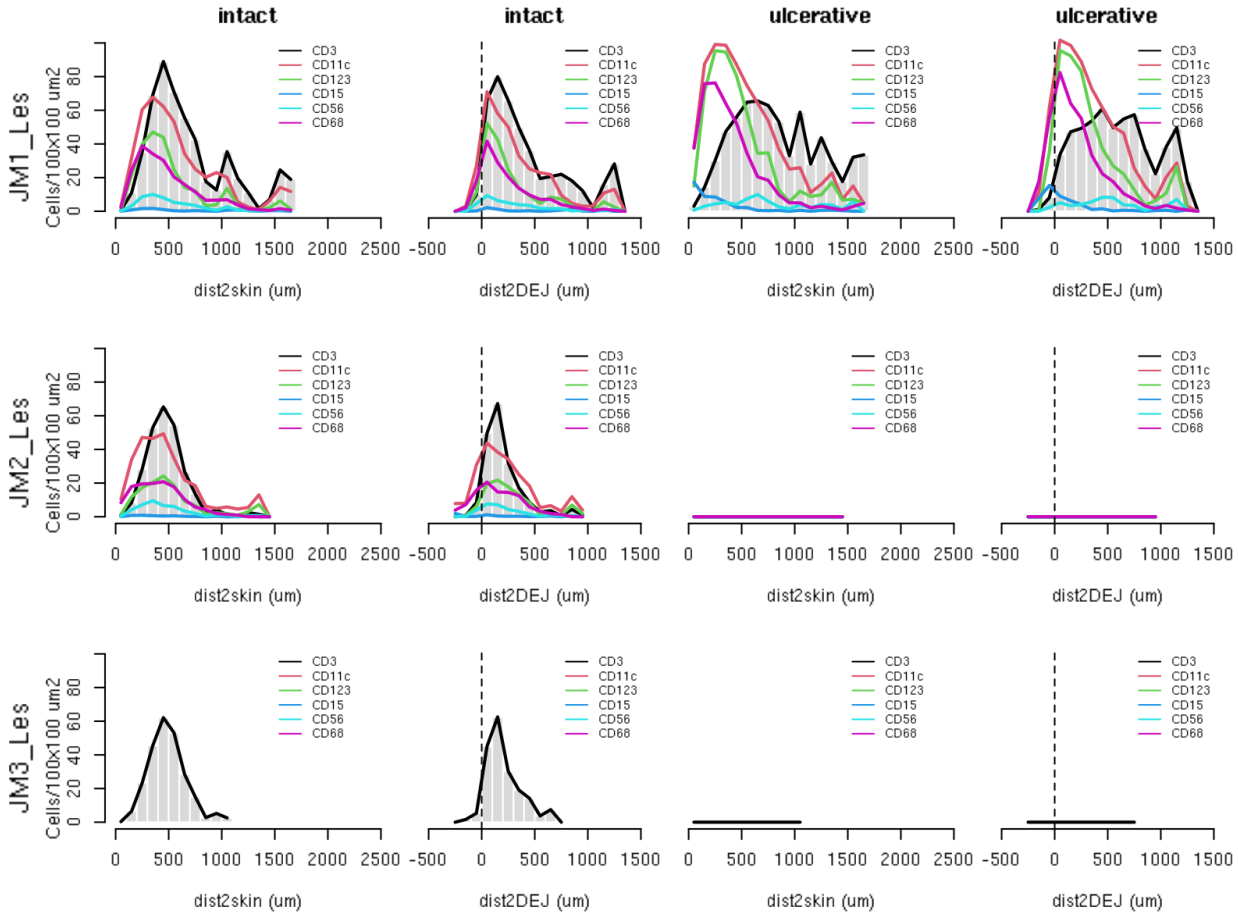

intact
intact
ulcerative
ulcerative
CF1_Les
CF2_Les
CF5_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

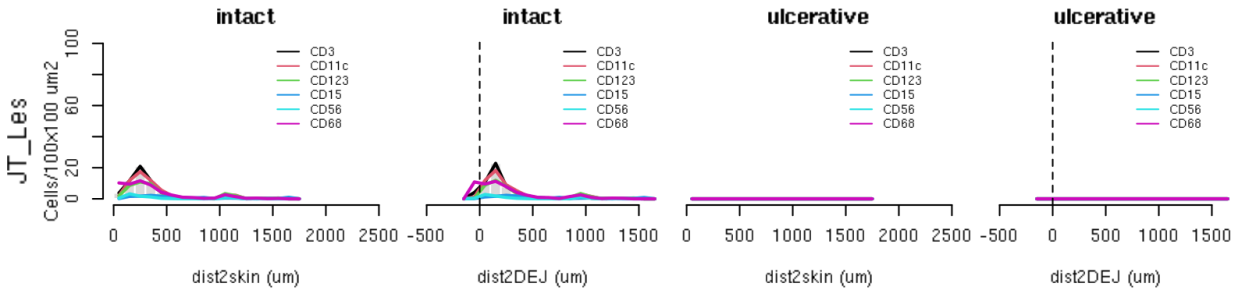

intact
intact
ulcerative
ulcerative
CJ1_Les
CJ2_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

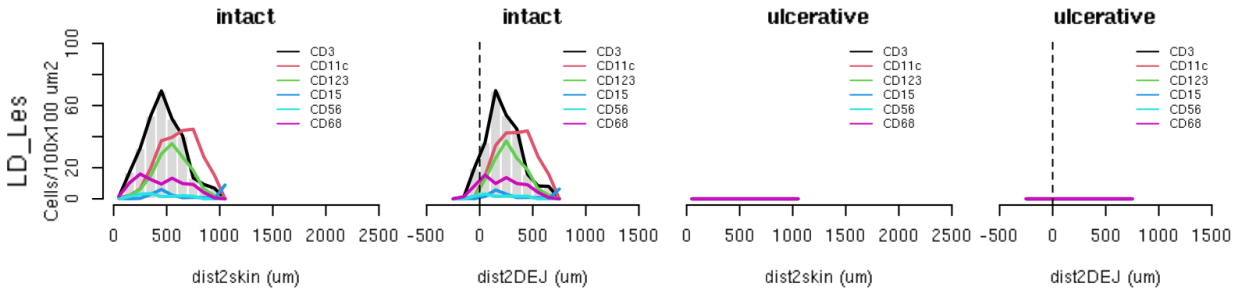

intact
intact
ulcerative
ulcerative
DD1_Les
DD2_Les
DD3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

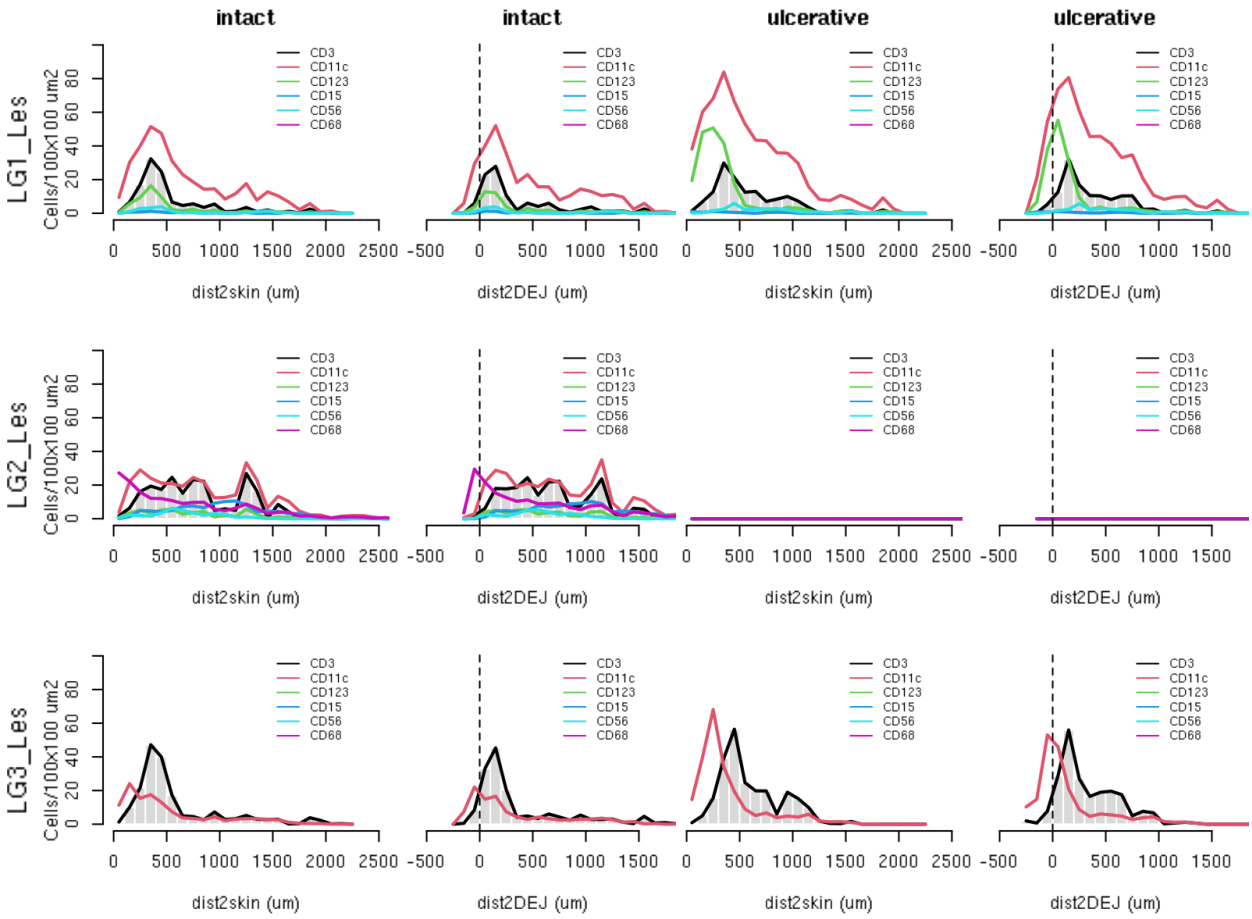

intact
intact
ulcerative
ulcerative
EB_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)

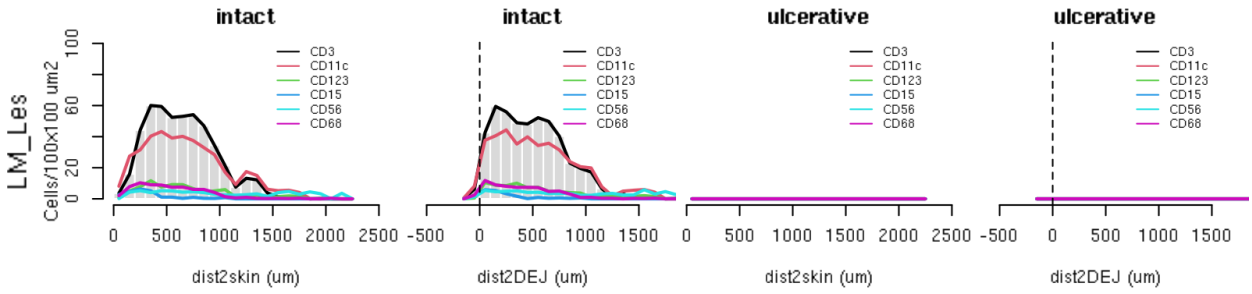

intact
intact
ulcerative
ulcerative
JK1_Les
JK2_Les
JK3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

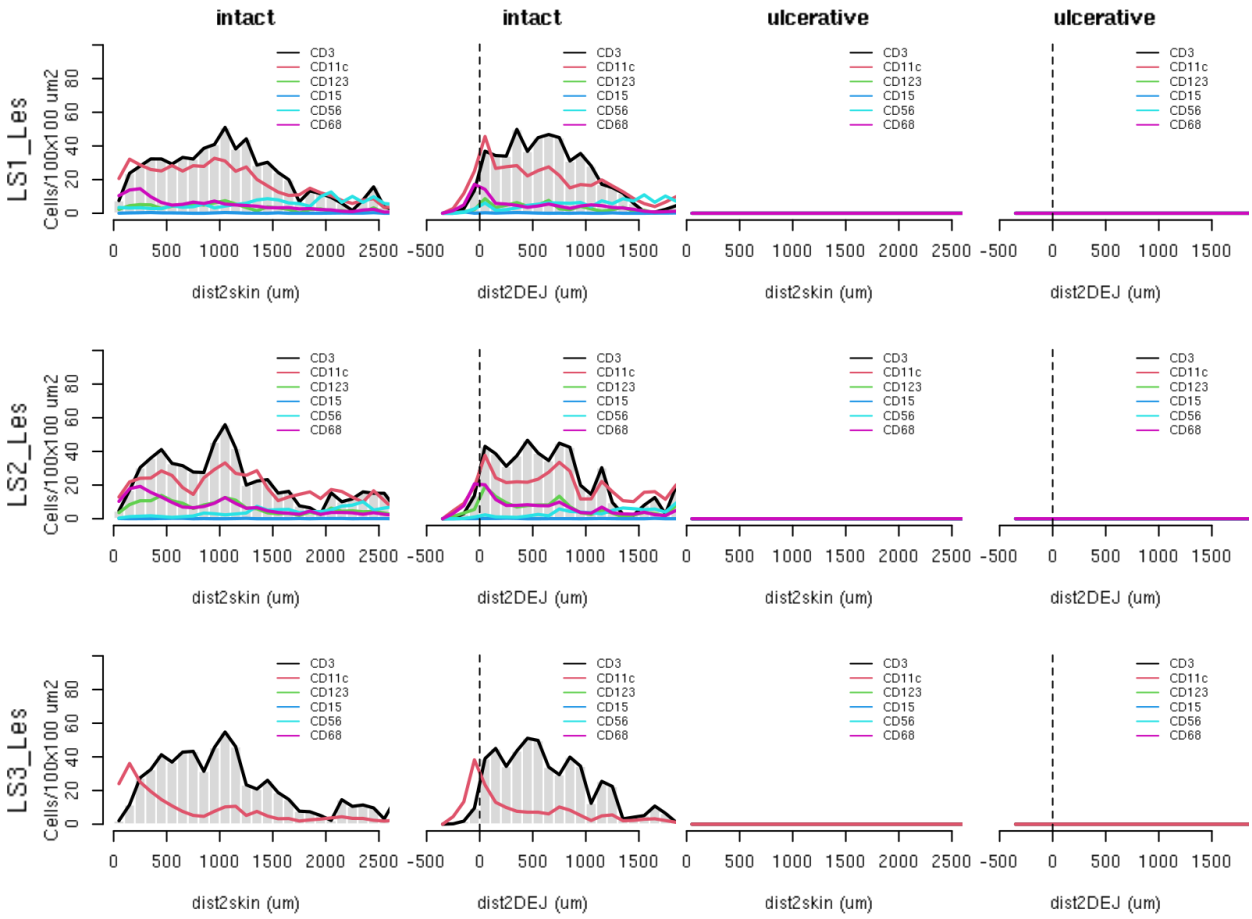

intact
intact
ulcerative
ulcerative
JM1_Les
JM2_Les
JM3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

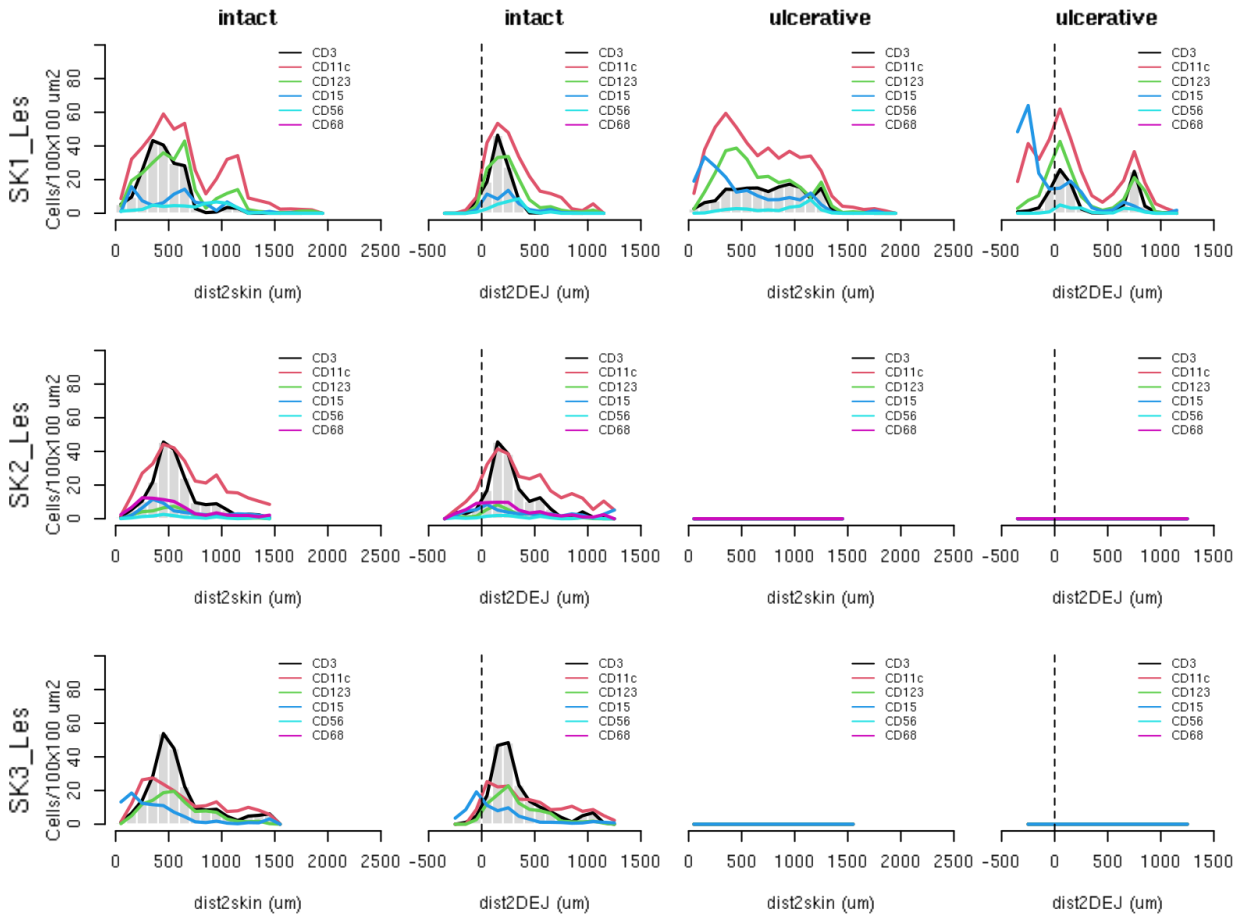

intact
intact
ulcerative
ulcerative
JT_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LD_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LG1_Les
LG2_Les
LG3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LM_Les
Cells/100x100 um2
0
20
60
100
CD3
CD11c
CD123
CD15
CD56
CD68
0
500
1000
1500
2000
2500
-500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
LS1_Les
LS2_Les
LS3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)


intact
intact
ulcerative
ulcerative
SK1_Les
SK2_Les
SK3_Les
Cells/100x100 um2
0 20 40 60 80
CD3
CD11c
CD123
CD15
CD56
CD68
0 500 1000 1500 2000 2500
-500 0 500 1000 1500
dist2skin (um)
dist2DEJ (um)

Count tables

For each slide, there is one table for up to 1000 um below DEJ.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	18	1372	0	0	90	1575			92	11968	0	0
CD123	15	1353	0	0	75	1553			76	11401	0	0
CD15	2	195	0	0	10	223			10	1887	0	0
CD279	69	1453	0	0	348	1668			354	15023	0	0
CD3	187	4071	0	0	944	4673			959	40233	0	0
CD4	92	3237	0	0	464	3716			472	32869	0	0
CD45	337	6773	0	0	1701	7775			2032	61578	0	0
CD56	12	363	0	0	60	416			61	3000	0	0
CD68	134	1037	0	0	676	1190			1900	6687	0	0
CD68_raw	247	933	0	0	1246	1071			2177	5710	0	0
CD69	233	1917	0	0	1176	2200			1195	18541	0	0
CD8	114	1343	0	0	575	1541			585	13245	0	0
Foxp3	6	93	0	0	30	106			30	860	0	0
Ki67	6	41	0	0	30	47			30	439	0	0
CD4T	51	2145	0	0	257	2462			261	23302	0	0
CD8T	119	1118	0	0	600	1283			610	10451	0	0
Treg	6	102	0	0	30	117			30	1013	0	0
area	0.20	0.87	0.00	0.00								

Table 21: AS1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1396	7069	0	0	2052	3499			3692	19029	0	0
CD123	605	3119	0	0	889	1543			1535	7817	0	0
CD279	624	3994	0	0	917	1976			2279	11773	0	0
CD3	1284	9955	0	0	1887	4927			3525	28840	0	0
CD4	1635	7638	0	0	2403	3780			6013	22417	0	0
CD4_raw	2081	8021	0	0	3059	3970			6838	22527	0	0
CD8	819	3921	0	0	1204	1940			2304	10928	0	0
Foxp3	134	423	0	0	197	209			509	1069	0	0
CD4T	390	5775	0	0	573	2858			1075	17528	0	0
CD8T	773	3216	0	0	1136	1591			2238	8815	0	0
Treg	59	401	0	0	86	198			208	878	0	0
area	0.68	2.02	0.00	0.00								

Table 22: AS2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD279	327	3620	0	0	569	1857			742	11869	0	0
CD3	578	10861	0	0	1005	5574			1261	36827	0	0
CD4	302	9296	0	0	525	4771			662	33106	0	0
CD56	77	1416	0	0	133	726			197	4611	0	0
CD8	538	4858	0	0	936	2493			1166	15415	0	0
Foxp3	82	256	0	0	142	131			186	613	0	0
CD4T	121	6027	0	0	210	3093			252	23020	0	0
CD8T	365	3756	0	0	635	1927			789	11899	0	0
Treg	21	130	0	0	36	66			42	365	0	0
area	0.57	1.95	0.00	0.00								

Table 23: AS3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	631	7926	1505	4139	1443	6261	3886	5829	2907	47019	12230	38420
CD123	124	3900	407	1457	283	3080	1051	2052	590	23344	3150	10980
CD15	8	2460	934	403	18	1943	2412	567	43	15167	10037	2924
CD279	159	1622	59	1249	363	1281	152	1759	744	9212	469	11828
CD3	405	7078	277	5041	926	5591	715	7100	1787	41150	2077	48287
CD4	334	5588	216	3488	764	4414	557	4912	1496	33455	1501	35234
CD45	540	8700	995	5538	1235	6872	2569	7800	2398	51401	7396	52348
CD56	64	529	36	404	146	417	92	569	271	2650	253	3475
CD68	1576	3700	0	0	3605	2922	0	0	7300	16390	0	0
CD8	296	3668	210	2990	677	2897	542	4211	2096	20629	1683	28301
Foxp3	42	348	82	222	96	274	211	312	257	2278	667	2579
Ki67	169	20	201	6	386	15	519	8	730	124	1677	88
CD4T	153	3029	55	2114	350	2392	142	2977	657	18086	376	21612
CD8T	180	3138	136	2860	411	2478	351	4028	782	17252	1041	26560
Treg	33	198	9	164	75	156	23	231	143	1167	60	1818
area	0.44	1.27	0.39	0.71								

Table 24: CF1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1018	9592	752	1646	1861	6216	2566	5611	5118	44861	8087	39279
CD123	292	7591	234	783	534	4919	798	2669	1518	32922	2387	14618
CD279	189	2016	152	516	345	1306	518	1758	895	8951	1455	10018
CD3	330	5949	303	1632	603	3855	1034	5563	1558	27734	2906	32435
CD4	136	4208	148	1017	248	2727	505	3466	695	21421	1420	28231
CD8	263	4535	261	1632	481	2939	890	5563	1156	21495	2443	37062
Foxp3	11	239	13	60	20	154	44	204	60	1137	144	1351
CD4T	73	1796	74	478	133	1164	252	1629	370	8976	706	15464
CD8T	148	2589	153	1043	270	1678	522	3555	622	11786	1429	21966
Treg	12	68	13	48	21	44	44	163	93	307	121	1028
area	0.55	1.54	0.29	0.29								

Table 25: CF2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	698	3776	0	0	1901	1970			2786	10618	0	0
CD15	614	3611	0	0	1672	1884			2841	13052	0	0
CD279	85	2577	0	0	231	1344			348	9604	0	0
CD3	263	5572	0	0	716	2907			1055	22191	0	0
CD4	505	10867	0	0	1375	5670			2738	45293	0	0
CD45	1514	12844	0	0	4123	6701			6687	50171	0	0
CD56	1	279	0	0	2	145			3	1263	0	0
CD68	1576	3700	0	0	4292	1930			8352	11096	0	0
CD69	256	2961	0	0	697	1545			1371	11209	0	0
CD8	200	1943	0	0	544	1013			1037	7934	0	0
Foxp3	0	8	0	0	0	4			0	34	0	0
Ki67	481	245	0	0	1310	127			4235	596	0	0
CD4T	112	4143	0	0	305	2161			438	17004	0	0
CD8T	96	1376	0	0	261	717			398	5583	0	0
area	0.37	1.92	0.00	0.00								

Table 26: CF5_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	157	3201	0	0	306	1975			394	15202	0	0
CD123	13	1440	0	0	25	888			32	6833	0	0
CD15	1	88	0	0	1	54			2	377	0	0
CD279	141	841	0	0	275	518			403	3900	0	0
CD3	196	2168	0	0	382	1337			539	11192	0	0
CD4	397	4852	0	0	775	2994			1145	21972	0	0
CD45	529	7523	0	0	1033	4642			1466	35142	0	0
CD56	14	866	0	0	27	534			34	4280	0	0
CD68	581	2248	0	0	1134	1387			1789	9939	0	0
CD68_raw	263	1226	0	0	513	756			834	5560	0	0
CD69	194	1736	0	0	378	1071			576	8617	0	0
CD8	44	779	0	0	85	480			115	4134	0	0
Foxp3	9	183	0	0	17	112			43	872	0	0
Ki67	257	91	0	0	501	56			683	461	0	0
CD4T	122	1540	0	0	238	950			336	7889	0	0
CD8T	35	602	0	0	68	371			86	3136	0	0
Treg	6	83	0	0	11	51			14	334	0	0
area	0.51	1.62	0.00	0.00								

Table 27: CJ1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	323	4247	0	0	494	2246			662	16131	0	0
CD123	56	1739	0	0	85	919			110	6993	0	0
CD15	3	28	0	0	4	14			5	99	0	0
CD3	36	2392	0	0	55	1265			80	10534	0	0
CD4	60	2626	0	0	91	1388			143	11469	0	0
CD45	489	6667	0	0	749	3526			1035	26158	0	0
CD68	399	2450	0	0	611	1295			862	8827	0	0
CD8	46	805	0	0	70	425			115	3640	0	0
CD4T	13	1172	0	0	19	619			35	5158	0	0
CD8T	8	583	0	0	12	308			15	2638	0	0
area	0.65	1.89	0.00	0.00								

Table 28: CJ2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	360	2203	10865	17752	3096	6651	8365	7851	7205	44740	43754	56722
CD123	308	1599	3062	14146	2649	4827	2357	6256	6258	28289	9729	44130
CD15	12	213	9301	3522	103	643	7161	1557	2180	3033	39371	8483
CD279	17	594	288	3275	146	1793	221	1448	318	12584	1015	11968
CD3	85	1535	140	6126	731	4634	107	2709	1741	30930	489	21917
CD4	233	2127	443	11429	2004	6421	341	5054	4824	42115	1359	40174
CD45	451	2935	12474	20869	3879	8861	9604	9230	11376	60192	50764	67113
CD56	7	134	3	612	60	404	2	270	131	3401	8	2272
CD68	141	673	1818	8087	1212	2031	1399	3576	3040	13207	6153	25334
CD69	19	253	31	2283	163	763	23	1009	356	5665	141	8652
CD8	37	634	75	2692	318	1914	57	1190	745	14085	254	9868
Foxp3_2	5	75	109	786	43	226	83	347	93	1879	485	3019
Foxp3	5	75	976	810	43	226	751	358	93	1879	3313	3019
CD4T	29	1160	21	4381	249	3502	16	1937	548	24519	64	16202
CD8T	37	474	22	1919	318	1431	16	848	749	10073	71	7024
Treg	6	80	1	628	51	241	0	277	112	1700	2	2356
area	0.12	0.33	1.30	2.26								

Table 29: DD1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	875	1740	1003	7548	3036	2783	1294	2853	7053	22057	1576	20163
CD123	266	743	759	5397	923	1188	979	2040	2279	6978	1460	8894
CD15	12	395	2932	3889	41	631	3784	1470	139	3852	16106	3487
CD279	27	40	37	128	93	63	47	48	226	421	65	280
CD3	293	2213	449	6549	1016	3540	579	2475	2662	22916	844	13175
CD4	241	1276	313	4330	836	2041	404	1636	2055	13312	529	9541
CD45	1148	4987	3613	22440	3984	7979	4664	8482	10267	54768	9448	48032
CD56	30	88	72	449	104	140	92	169	264	899	112	1131
CD68	162	249	413	1451	562	398	533	548	1371	2149	1218	2825
CD69	530	1530	578	5345	1839	2447	746	2020	4454	17518	898	10898
CD8	124	615	129	1683	430	983	166	636	1051	5981	208	3520
Foxp3	31	21	21	140	107	33	27	52	383	240	32	322
Ki67	38	20	729	549	131	31	941	207	787	281	3092	367
CD4T	94	702	134	2163	326	1123	172	817	850	7981	207	4803
CD8T	109	448	110	1174	378	716	142	443	885	4260	171	2451
Treg	25	5	18	68	86	7	23	25	252	82	28	132
area	0.29	0.63	0.77	2.65								

Table 30: DD2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	478	2859	10260	17020	2745	6305	8695	6913	5194	42397	47109	44446
CD123	458	2163	3003	13393	2630	4770	2545	5440	4761	27254	7639	29442
CD15	46	529	8776	3516	264	1166	7437	1428	1073	6368	41005	5953
CD279	22	258	186	1580	126	568	157	641	237	4698	449	4312
CD3	97	1086	355	3700	557	2395	300	1502	1080	17451	799	10727
CD4	372	2498	598	12097	2136	5509	506	4913	3918	36737	1312	35212
CD45	509	3197	11142	17714	2923	7050	9442	7195	5589	46301	52430	46066
CD56	18	159	41	669	103	350	34	271	208	2951	84	2102
CD68	344	1487	2300	10029	1975	3279	1949	4073	5034	22858	7049	26343
CD69	2	63	96	388	11	138	81	157	28	1208	198	1215
CD8	61	722	211	2741	350	1592	178	1113	691	12632	514	8445
Foxp3	14	47	219	298	80	103	185	121	178	851	573	709
CD4T	51	854	221	2939	292	1883	187	1193	550	14457	462	8858
CD8T	34	391	148	1351	195	862	125	548	414	6706	314	4069
Treg	5	19	24	135	28	41	20	54	56	324	49	400
area	0.17	0.45	1.18	2.46								

Table 31: DD3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	56	5795	0	0	177	3802			257	32673	0	0
CD123	47	3656	0	0	148	2398			199	19758	0	0
CD15	9	178	0	0	28	116			36	793	0	0
CD279	45	203	0	0	142	133			181	1074	0	0
CD3	143	8872	0	0	452	5820			575	50879	0	0
CD4	221	10807	0	0	699	7090			1050	59526	0	0
CD45	391	14245	0	0	1236	9346			1615	77705	0	0
CD56	5	825	0	0	15	541			30	4592	0	0
CD68	121	4464	0	0	382	2928			551	24026	0	0
CD8	62	3844	0	0	196	2522			368	22116	0	0
Foxp3	9	69	0	0	28	45			36	390	0	0
Ki67	200	83	0	0	632	54			923	440	0	0
CD4T	74	5830	0	0	234	3825			297	33639	0	0
CD8T	36	2753	0	0	113	1806			144	15863	0	0
Treg	2	49	0	0	6	32			8	269	0	0
area	0.32	1.52	0.00	0.00								

Table 32: EB_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	255	3684	0	0	309	3277			585	18349	0	0
CD123	22	435	0	0	26	386			44	2053	0	0
CD15	74	1738	0	0	89	1546			1646	13393	0	0
CD279	22	97	0	0	26	86			63	310	0	0
CD3	26	391	0	0	31	347			50	1321	0	0
CD4	69	1659	0	0	83	1475			189	9044	0	0
CD56	6	148	0	0	7	131			11	643	0	0
CD68	1321	1568	0	0	1601	1394			2983	10585	0	0
CD68_raw	1701	2379	0	0	2062	2116			3900	14474	0	0
CD69	33	165	0	0	40	146			64	390	0	0
CD8	24	110	0	0	29	97			46	246	0	0
Foxp3	10	54	0	0	12	48			24	64	0	0
Ki67	669	48	0	0	811	42			1308	43	0	0
CD4T	8	249	0	0	9	221			15	828	0	0
CD8T	18	74	0	0	21	65			34	164	0	0
Treg	3	25	0	0	3	22			5	19	0	0
area	0.82	1.12	0.00	0.00								

Table 33: JK1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	18	417	0	0	23	373			32	2049	0	0
CD4	86	1748	0	0	112	1563			216	7623	0	0
CD8	13	97	0	0	17	86			27	269	0	0
CD4T	7	271	0	0	9	242			13	1216	0	0
CD8T	10	62	0	0	13	55			19	175	0	0
area	0.76	1.12	0.00	0.00								

Table 34: JK2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD123	17	459	0	0	23	462			36	3431	0	0
CD15	7	717	0	0	9	722			19	6137	0	0
CD279	17	146	0	0	23	147			35	766	0	0
CD3	20	251	0	0	27	253			42	1295	0	0
CD4	68	1422	0	0	94	1433			154	5903	0	0
CD68	756	1795	0	0	1045	1809			1882	8426	0	0
CD68_raw	1263	1499	0	0	1745	1511			3015	7592	0	0
CD69	22	158	0	0	30	159			91	1070	0	0
CD8	15	56	0	0	20	56			31	169	0	0
Foxp3	5	35	0	0	6	35			57	212	0	0
Ki67	158	2	0	0	218	2			342	3	0	0
CD4T	5	176	0	0	6	177			10	974	0	0
CD8T	12	31	0	0	16	31			25	78	0	0
Treg	2	17	0	0	2	17			4	161	0	0
area	0.72	0.99	0.00	0.00								

Table 35: JK3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	929	5747	1199	7120	1912	5058	4988	8169	2992	22825	7293	46629
CD123	376	3507	649	5802	773	3086	2700	6656	1161	11487	3566	34103
CD15	6	122	326	345	12	107	1356	395	22	398	2312	1437
CD279	182	2440	62	1946	374	2147	257	2232	568	8567	337	14251
CD3	336	6967	155	4928	691	6131	644	5654	1067	32393	916	43238
CD4	621	7178	448	5687	1278	6317	1863	6525	1983	32235	2730	45746
CD45	1238	11869	1638	10638	2548	10446	6814	12205	4000	61369	11128	84620
CD56	152	670	44	495	312	589	183	567	465	2431	239	3978
CD68	623	2813	1122	4096	1282	2475	4667	4699	2008	9515	7316	21477
CD68_raw	860	2235	931	3089	1770	1967	3873	3544	2799	7405	6400	16275
CD69	731	7062	898	7673	1504	6215	3736	8803	2271	26876	5068	51526
CD8	176	2236	62	1565	362	1967	257	1795	550	9114	349	13401
Foxp3	32	257	9	126	65	226	37	144	112	1026	48	923
Ki67	118	54	66	63	242	47	274	72	362	318	653	387
CD4T	140	4534	54	3137	288	3990	224	3599	448	21403	293	28024
CD8T	154	1840	42	1263	316	1619	174	1449	487	7089	228	10450
Treg	30	298	12	178	61	262	49	204	102	1063	65	1187
area	0.49	1.14	0.24	0.87								

Table 36: JM1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1561	4294	0	0	2517	4465			4654	14702	0	0
CD123	298	2103	0	0	480	2187			678	7352	0	0
CD15	12	76	0	0	19	79			228	219	0	0
CD279	161	511	0	0	259	531			362	1322	0	0
CD3	403	4870	0	0	649	5064			912	13728	0	0
CD4	656	5371	0	0	1058	5585			1510	20365	0	0
CD45	798	4536	0	0	1287	4717			1918	12324	0	0
CD56	188	678	0	0	303	705			425	1876	0	0
CD68	848	1857	0	0	1367	1931			2693	5393	0	0
CD68_raw	2839	2193	0	0	4578	2280			8979	6295	0	0
CD69	791	6050	0	0	1275	6291			1766	17855	0	0
CD8	211	2087	0	0	340	2170			460	5812	0	0
Foxp3	52	450	0	0	83	467			120	1352	0	0
Ki67	981	135	0	0	1582	140			2166	236	0	0
CD4T	195	2900	0	0	314	3015			440	8648	0	0
CD8T	120	1214	0	0	193	1262			261	2592	0	0
Treg	38	243	0	0	61	252			85	600	0	0
area	0.62	0.96	0.00	0.00								

Table 37: JM2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	233	4303	0	0	449	5757			692	13715	0	0
CD8	170	1750	0	0	328	2341			511	5158	0	0
Foxp3	31	113	0	0	59	151			97	270	0	0
CD8T	94	1192	0	0	181	1594			267	2942	0	0
area	0.52	0.75	0.00	0.00								

Table 38: JM3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	24	1182	0	0	107	781			107	3843	0	0
CD123	9	896	0	0	40	592			40	3075	0	0
CD15	4	218	0	0	17	144			17	943	0	0
CD279	15	149	0	0	66	98			66	522	0	0
CD3	92	1195	0	0	410	790			410	4067	0	0
CD4	72	1189	0	0	321	786			321	4144	0	0
CD45	161	3393	0	0	718	2243			718	12420	0	0
CD56	8	174	0	0	35	115			35	467	0	0
CD68	243	923	0	0	1084	610			1080	3126	0	0
CD68_raw	558	956	0	0	2489	632			2490	3296	0	0
CD8	54	518	0	0	240	342			241	2000	0	0
Foxp3	70	97	0	0	312	64			312	335	0	0
Ki67	77	5	0	0	343	3			343	19	0	0
CD4T	44	664	0	0	196	439			196	2267	0	0
CD8T	43	342	0	0	191	226			191	1225	0	0
Treg	2	75	0	0	8	49			8	239	0	0
area	0.22	1.51	0.00	0.00								

Table 39: JT_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	41	2501	0	0	118	4100			212	20605	0	0
CD123	20	1599	0	0	58	2621			74	11693	0	0
CD15	5	176	0	0	14	288			48	1762	0	0
CD279	89	440	0	0	258	721			381	2401	0	0
CD3	540	3111	0	0	1566	5100			2094	20040	0	0
CD4	209	2014	0	0	606	3302			829	14072	0	0
CD45	794	5665	0	0	2302	9288			3180	39615	0	0
CD56	48	144	0	0	139	236			179	689	0	0
CD68	233	862	0	0	675	1413			968	4728	0	0
CD69	392	2232	0	0	1136	3659			1512	16013	0	0
CD8	223	993	0	0	646	1628			832	5963	0	0
Foxp3	14	42	0	0	40	68			72	245	0	0
Ki67	57	1392	0	0	165	2282			460	16394	0	0
CD4T	90	729	0	0	261	1195			375	5366	0	0
CD8T	183	410	0	0	530	672			682	1979	0	0
Treg	8	13	0	0	23	21			29	71	0	0
area	0.34	0.61	0.00	0.00								

Table 40: LD_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1074	5057	942	5215	2325	3056	4586	5237	3520	19296	7543	37327
CD123	132	916	613	1498	285	553	2984	1504	405	2448	4437	5717
CD15	6	101	14	49	12	61	68	49	23	320	143	301
CD279	67	339	13	257	145	204	63	258	206	813	84	2144
CD3	205	2000	83	1327	443	1208	404	1332	610	6164	554	10356
CD4	303	2587	95	2189	656	1563	462	2198	905	8683	672	17517
CD45	967	6473	798	5806	2093	3912	3885	5830	3051	24134	6186	43817
CD56	34	247	6	242	73	149	29	243	110	703	39	2022
CD69	169	1166	73	1225	365	704	355	1230	560	3303	496	9320
CD8	178	1663	69	1223	385	1005	335	1228	553	5818	463	9485
Foxp3	66	350	21	223	142	211	102	223	220	1048	137	1655
Ki67	34	36	1	22	73	21	4	22	110	173	6	188
CD4T	82	1003	25	764	177	606	121	767	245	3124	163	6189
CD8T	84	831	24	491	181	502	116	493	245	2800	156	3949
Treg	25	150	5	68	54	90	24	68	73	306	32	497
area	0.46	1.65	0.21	1.00								

Table 41: LG1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	93	5966	0	0	274	2430			348	18757	0	0
CD123	5	1047	0	0	14	426			15	3282	0	0
CD15	1	1619	0	0	2	659			3	5904	0	0
CD279	43	537	0	0	126	218			132	1727	0	0
CD3	58	4379	0	0	170	1783			178	15120	0	0
CD4	92	5078	0	0	271	2068			348	16951	0	0
CD45	194	9993	0	0	571	4070			659	32381	0	0
CD56	8	874	0	0	23	355			24	2897	0	0
CD68	962	3235	0	0	2836	1317			3281	8861	0	0
CD68_raw	967	1731	0	0	2850	705			3041	4593	0	0
CD69	98	2628	0	0	288	1070			301	8514	0	0
CD8	40	2504	0	0	117	1019			123	8530	0	0
Foxp3	3	110	0	0	8	44			9	372	0	0
Ki67	293	50	0	0	863	20			902	158	0	0
CD4T	29	3053	0	0	85	1243			89	10706	0	0
CD8T	30	1958	0	0	88	797			92	6725	0	0
Treg	1	61	0	0	2	24			3	210	0	0
area	0.34	2.46	0.00	0.00								

Table 42: LG2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	287	2999	149	2526	632	1716	472	2489	886	9631	1032	17512
CD4	403	3396	241	3053	887	1944	763	3008	1227	12074	1447	21021
CD68	828	1401	1139	1512	1824	801	3608	1489	2943	4648	7797	6171
CD68_raw	1527	2002	1657	2178	3364	1146	5250	2146	5489	6722	11556	9982
CD8	187	1791	97	1647	411	1025	307	1622	584	5793	735	12035
CD4T	107	1730	51	1456	235	990	161	1434	323	6012	329	10194
CD8T	141	1413	71	1149	310	808	224	1132	441	4554	499	8184
area	0.45	1.75	0.32	1.01								

Table 43: LG3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	298	7681	0	0	745	4314			871	30612	0	0
CD123	18	1632	0	0	45	916			49	5907	0	0
CD15	65	517	0	0	162	290			179	1292	0	0
CD279	39	177	0	0	97	99			107	549	0	0
CD3	225	9871	0	0	563	5544			646	39738	0	0
CD4	354	16036	0	0	885	9006			1003	65640	0	0
CD45	482	18467	0	0	1206	10372			1427	77612	0	0
CD56	45	926	0	0	112	520			124	3538	0	0
CD68	88	1563	0	0	220	877			243	5272	0	0
CD69	77	2900	0	0	192	1628			213	12362	0	0
CD8	125	4096	0	0	312	2300			369	17290	0	0
Foxp3	6	142	0	0	15	79			16	697	0	0
CD4T	70	7909	0	0	175	4442			193	32370	0	0
CD8T	67	1663	0	0	167	934			208	6790	0	0
Treg	4	89	0	0	10	49			11	384	0	0
area	0.40	1.78	0.00	0.00								

Table 44: LM_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1402	7533	0	0	2030	3501			3949	21233	0	0
CD123	119	1401	0	0	172	651			334	4008	0	0
CD15	16	49	0	0	23	22			86	133	0	0
CD279	496	4097	0	0	718	1904			1321	14529	0	0
CD3	692	10338	0	0	1002	4804			1633	35819	0	0
CD4	813	10287	0	0	1177	4781			2016	33067	0	0
CD45	1692	14059	0	0	2450	6534			4327	45292	0	0
CD56	133	1209	0	0	192	561			312	4186	0	0
CD68	908	1720	0	0	1315	799			2368	4082	0	0
CD68_raw	1423	2347	0	0	2060	1090			3706	5522	0	0
CD69	1009	7850	0	0	1461	3648			2446	25027	0	0
CD8	291	3429	0	0	421	1593			692	11744	0	0
Foxp3	134	1145	0	0	194	532			328	3443	0	0
Ki67	359	54	0	0	519	25			831	167	0	0
CD4T	355	7865	0	0	514	3655			866	27713	0	0
CD8T	218	2375	0	0	315	1103			481	7895	0	0
Treg	108	862	0	0	156	400			266	2388	0	0
area	0.69	2.15	0.00	0.00								

Table 45: LS1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1567	7637	0	0	1490	3340			3296	21412	0	0
CD123	477	3198	0	0	453	1399			1030	7797	0	0
CD15	6	30	0	0	5	13			12	75	0	0
CD279	342	1547	0	0	325	676			603	4389	0	0
CD3	980	11077	0	0	932	4845			1687	33467	0	0
CD4	1229	12721	0	0	1168	5565			2143	37434	0	0
CD45	1941	15205	0	0	1845	6651			3742	45520	0	0
CD56	72	644	0	0	68	281			119	2287	0	0
CD68	1610	3072	0	0	1531	1343			3155	7119	0	0
CD68_raw	3474	4839	0	0	3303	2116			7336	11341	0	0
CD69	652	5312	0	0	620	2323			1119	15818	0	0
CD8	302	2926	0	0	287	1280			505	8709	0	0
Foxp3	112	612	0	0	106	267			217	1526	0	0
Ki67	542	31	0	0	515	13			910	100	0	0
CD4T	459	7812	0	0	436	3417			766	24434	0	0
CD8T	212	1837	0	0	201	803			348	5374	0	0
Treg	74	420	0	0	70	183			120	1009	0	0
area	1.05	2.29	0.00	0.00								

Table 46: LS2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	643	11896	0	0	646	5140			1152	36142	0	0
CD4	797	13016	0	0	800	5624			1507	37868	0	0
CD68	2794	3479	0	0	2807	1503			5603	7474	0	0
CD68_raw	4129	4884	0	0	4149	2110			8835	10839	0	0
CD8	319	3956	0	0	320	1709			540	11834	0	0
CD4T	315	6833	0	0	316	2952			580	20876	0	0
CD8T	192	2494	0	0	192	1077			326	7140	0	0
area	1.00	2.31	0.00	0.00								

Table 47: LS3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	84	2383	3133	4770	909	3323	3948	4900	1129	19527	13606	17775
CD123	37	1361	1582	2848	400	1898	1993	2925	425	10599	4810	8845
CD15	18	475	1775	1494	194	662	2237	1534	206	3411	15051	5021
CD279	43	791	545	1384	465	1103	686	1421	493	6510	1669	3561
CD3	69	1063	717	1686	746	1482	903	1732	792	8403	1889	5629
CD4	30	952	421	1960	324	1327	530	2013	344	8031	1068	5633
CD45	199	3631	4090	7433	2154	5064	5155	7636	2285	29107	17812	28517
CD56	6	262	41	380	64	365	51	390	68	2310	91	1337
CD8	36	461	315	965	389	642	397	991	413	3658	817	2221
Foxp3	13	145	107	132	140	202	134	135	149	969	468	131
CD4T	22	668	265	1114	238	931	334	1144	252	5768	621	3641
CD8T	22	668	265	1114	238	931	334	1144	252	5768	621	3641
Treg	11	169	76	117	119	235	95	120	126	1291	172	176
area	0.09	0.72	0.79	0.97								

Table 48: SK1_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	1024	7035	0	0	1491	3216			3213	21119	0	0
CD123	96	945	0	0	139	432			387	2447	0	0
CD15	344	1075	0	0	500	491			1185	2607	0	0
CD279	136	827	0	0	198	378			406	2194	0	0
CD3	343	4600	0	0	499	2103			1266	13609	0	0
CD4	580	5479	0	0	844	2505			1811	15880	0	0
CD45	758	7130	0	0	1103	3259			2406	21562	0	0
CD56	32	281	0	0	46	128			136	782	0	0
CD68	551	1528	0	0	802	698			1764	3957	0	0
CD69	744	3693	0	0	1083	1688			2522	9820	0	0
CD8	146	914	0	0	212	417			482	2578	0	0
Foxp3	42	222	0	0	61	101			177	611	0	0
Ki67	210	24	0	0	305	10			443	14	0	0
CD4T	148	3250	0	0	215	1485			584	9848	0	0
CD8T	80	630	0	0	116	288			269	1775	0	0
Treg	69	433	0	0	100	197			298	926	0	0
area	0.69	2.19	0.00	0.00								

Table 49: SK2_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.

	Counts				Density				normalized Density			
	intact		ulcerative		intact		ulcerative		intact		ulcerative	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD11c	444	4568	0	0	740	1967			891	12313	0	0
CD123	135	2885	0	0	225	1242			268	8055	0	0
CD279	34	124	0	0	56	53			67	314	0	0
CD3	258	5477	0	0	430	2359			599	16013	0	0
CD4	355	5541	0	0	592	2386			790	15836	0	0
CD68	1047	1423	0	0	1746	613			3140	3008	0	0
CD68_raw	1946	3284	0	0	3245	1414			5326	7196	0	0
CD8	123	1305	0	0	205	562			279	3601	0	0
Foxp3	38	300	0	0	63	129			101	782	0	0
CD4T	115	3311	0	0	191	1426			263	9825	0	0
CD8T	106	800	0	0	176	344			236	2086	0	0
Treg	28	199	0	0	46	85			55	491	0	0
area	0.60	2.32	0.00	0.00								

Table 50: SK3_Les. All measurements are up to 1000 um below DEJ. Area is measured in mm2. Density in count per mm2. normalized density refers to the average of positive density measurements taken at 100um DEJ depth intervals, i.e. each depth is weighted equally instead of based on tissue shape.